

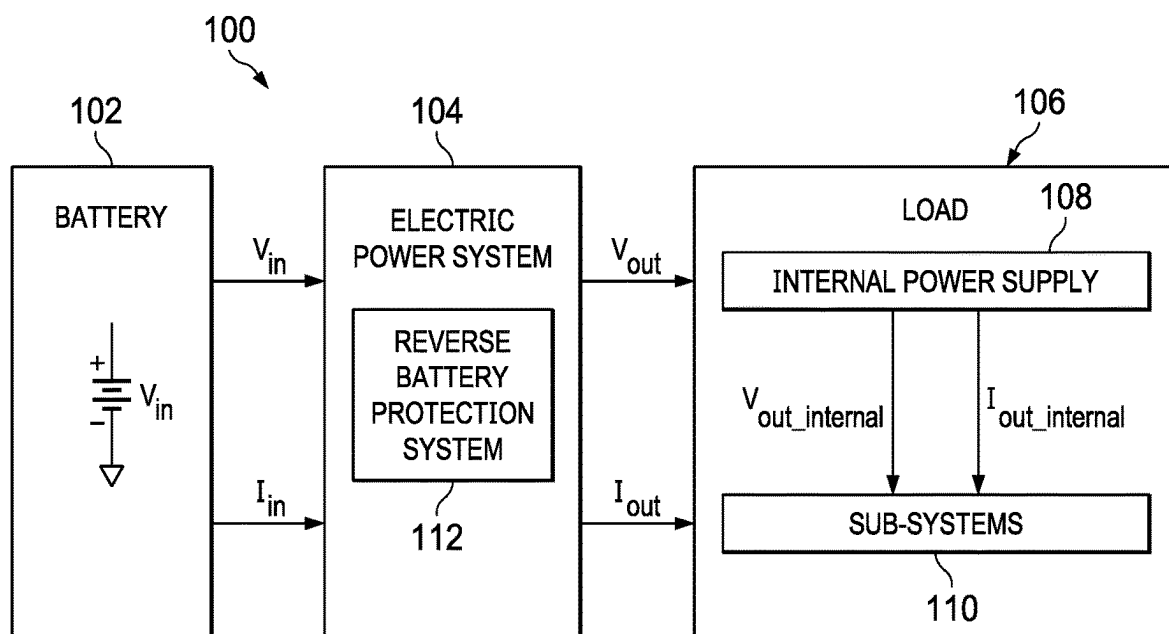


FIG. 1

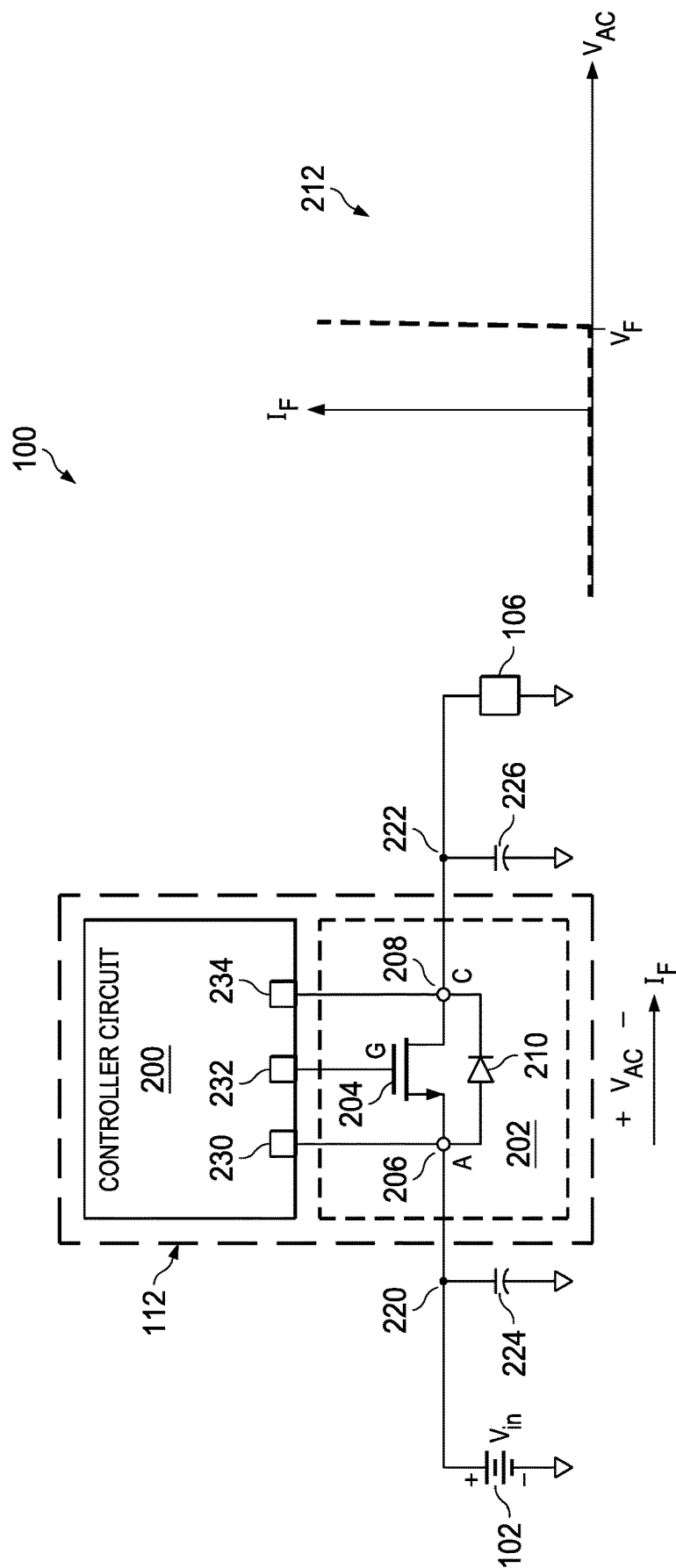
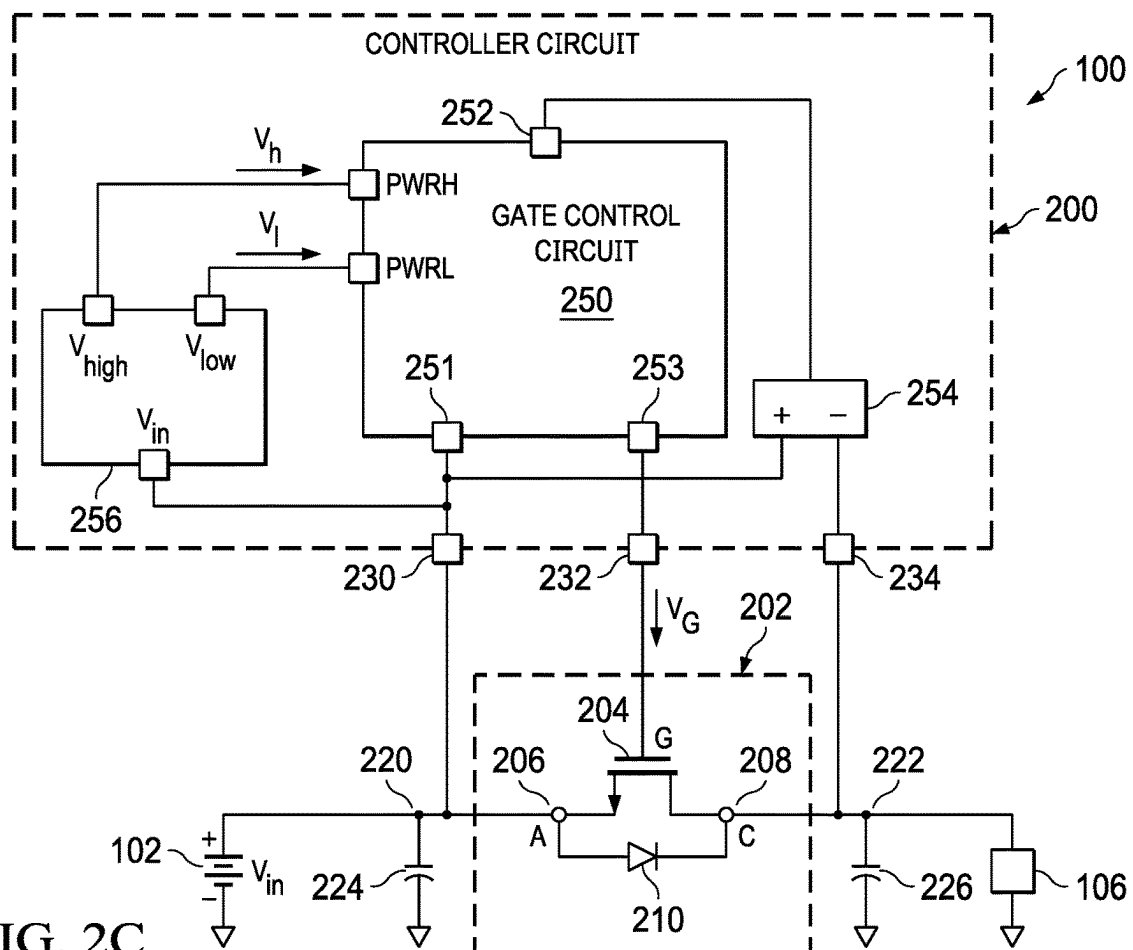
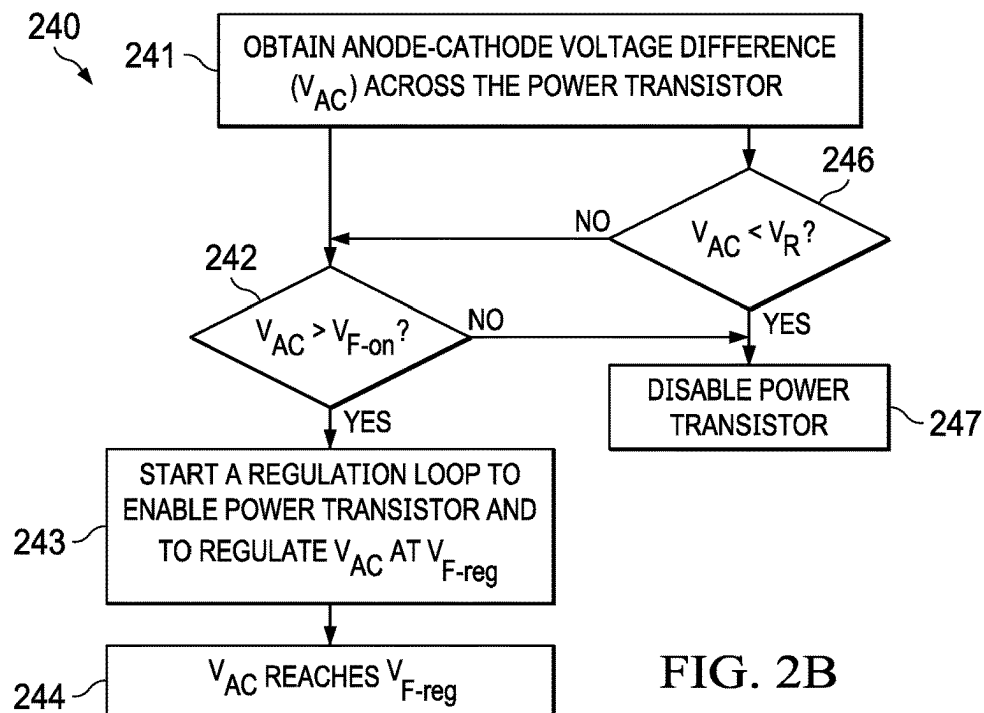


FIG. 2A



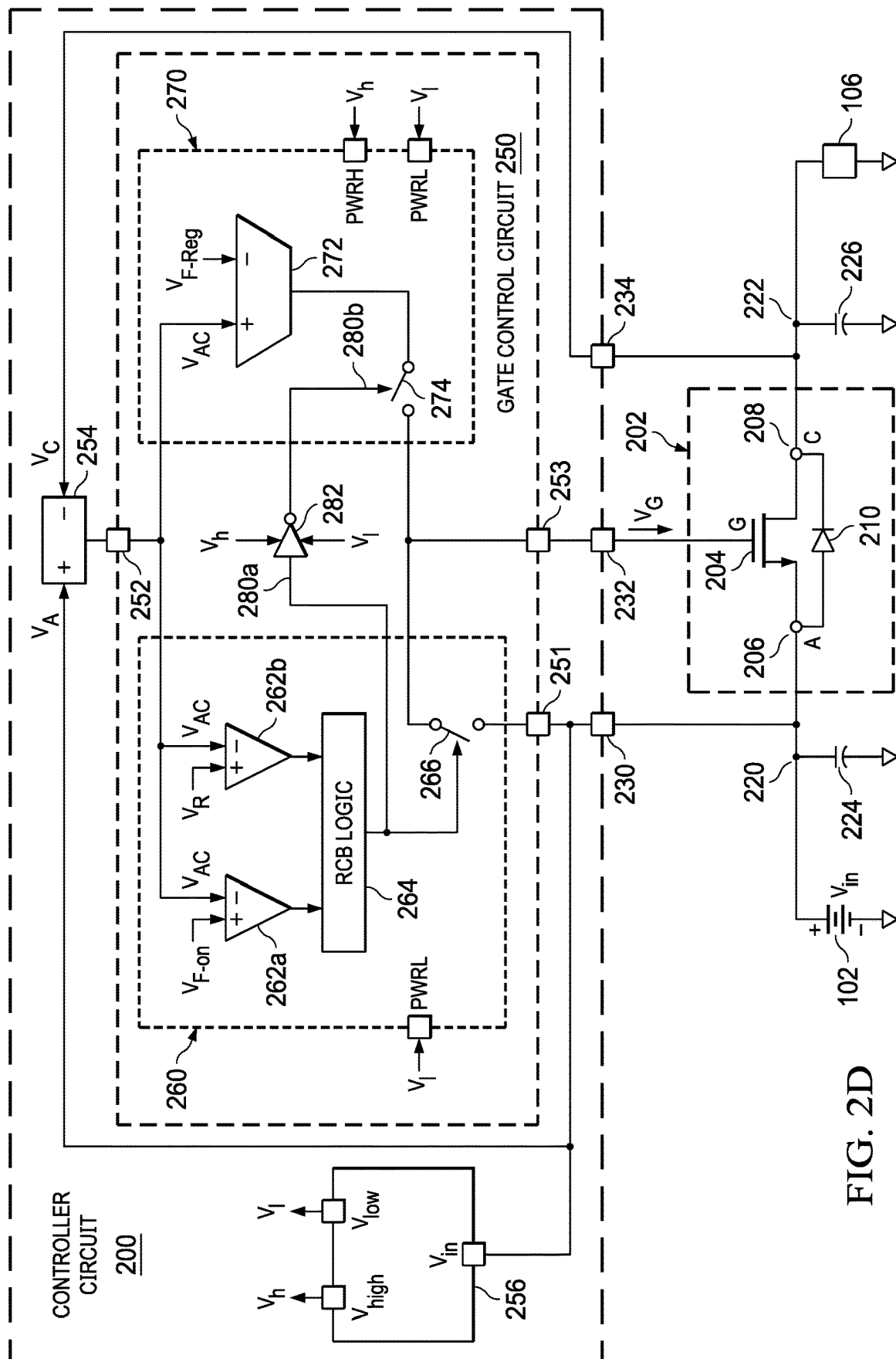


FIG. 2D

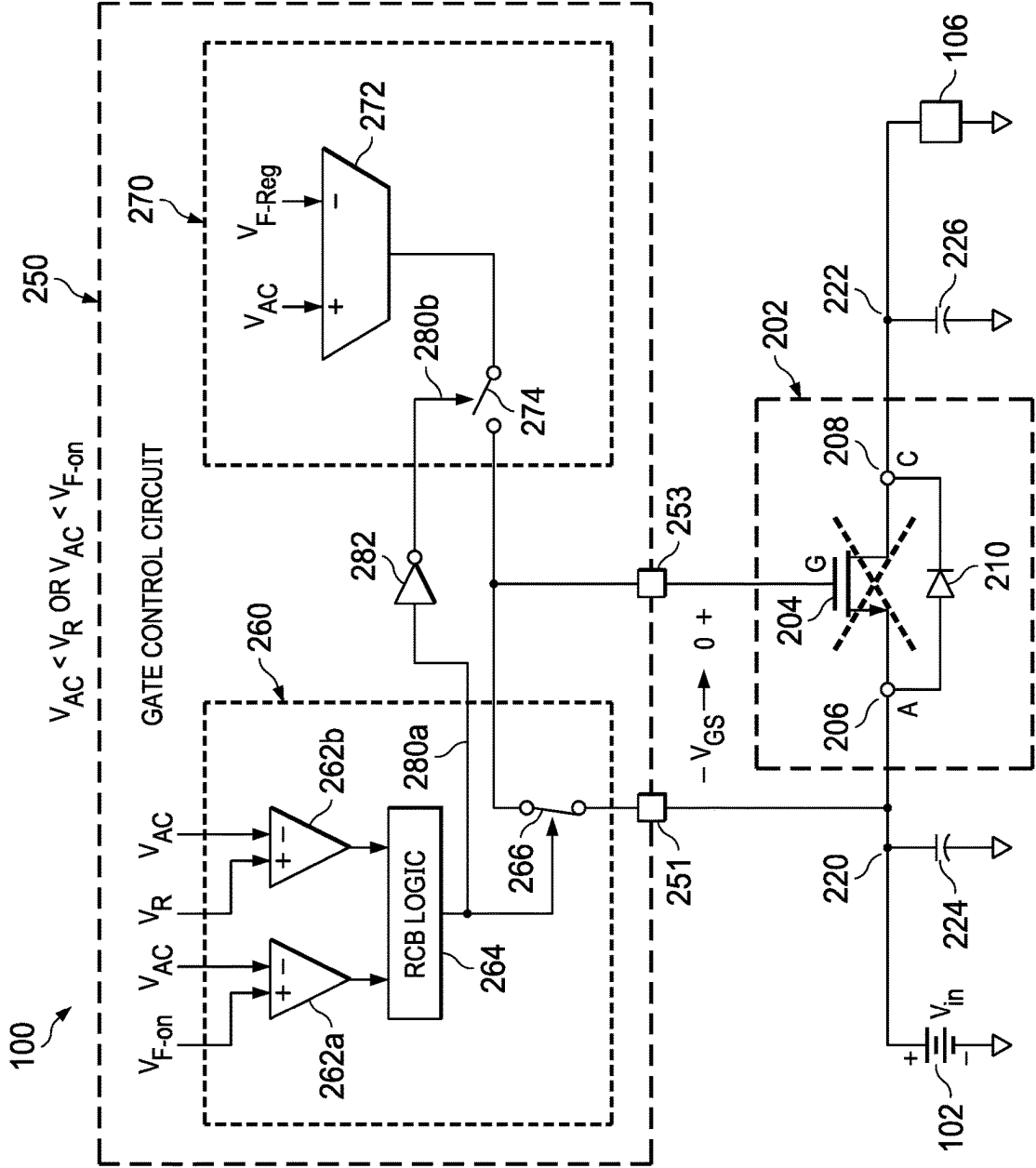


FIG. 2E

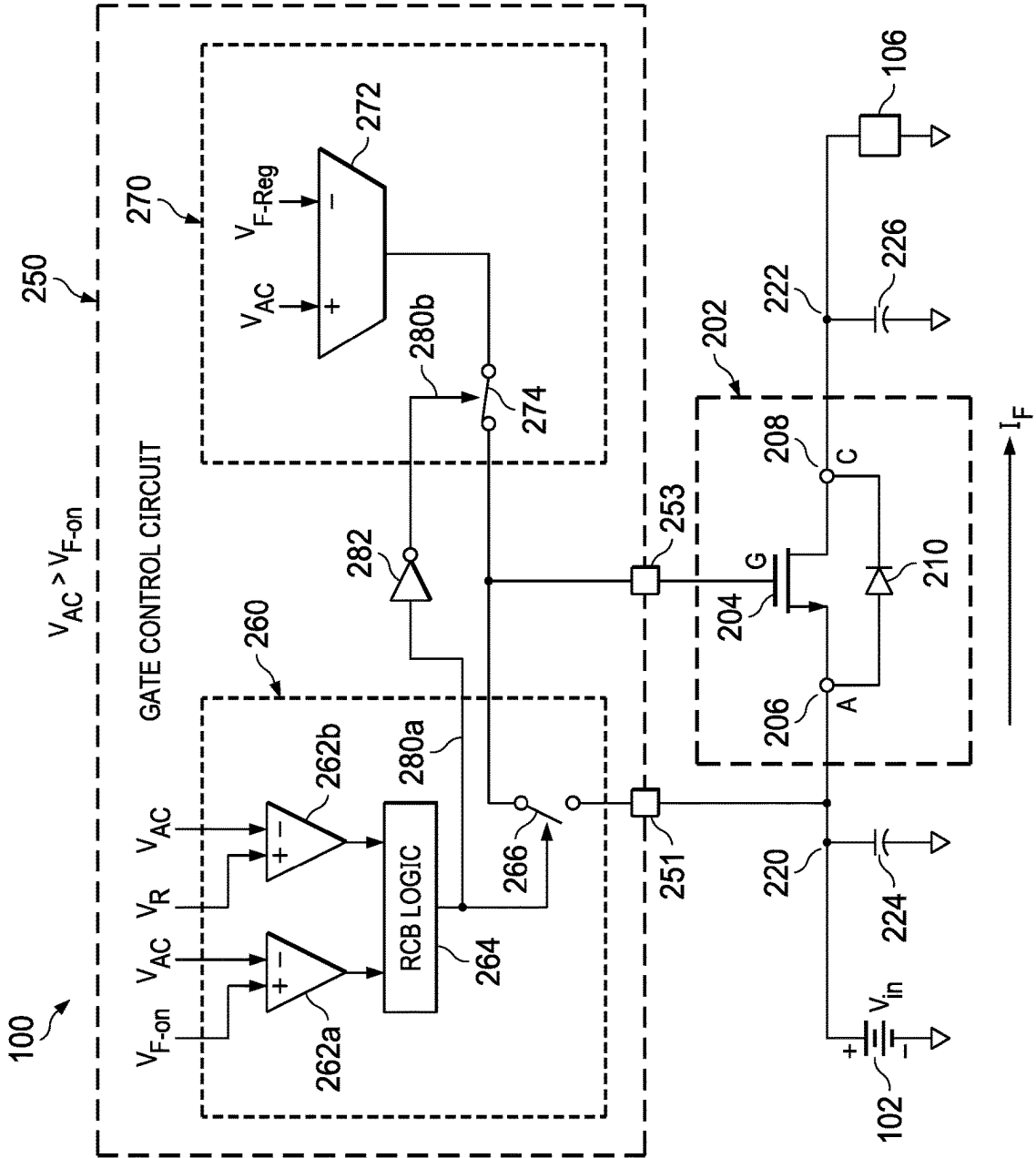


FIG. 2F

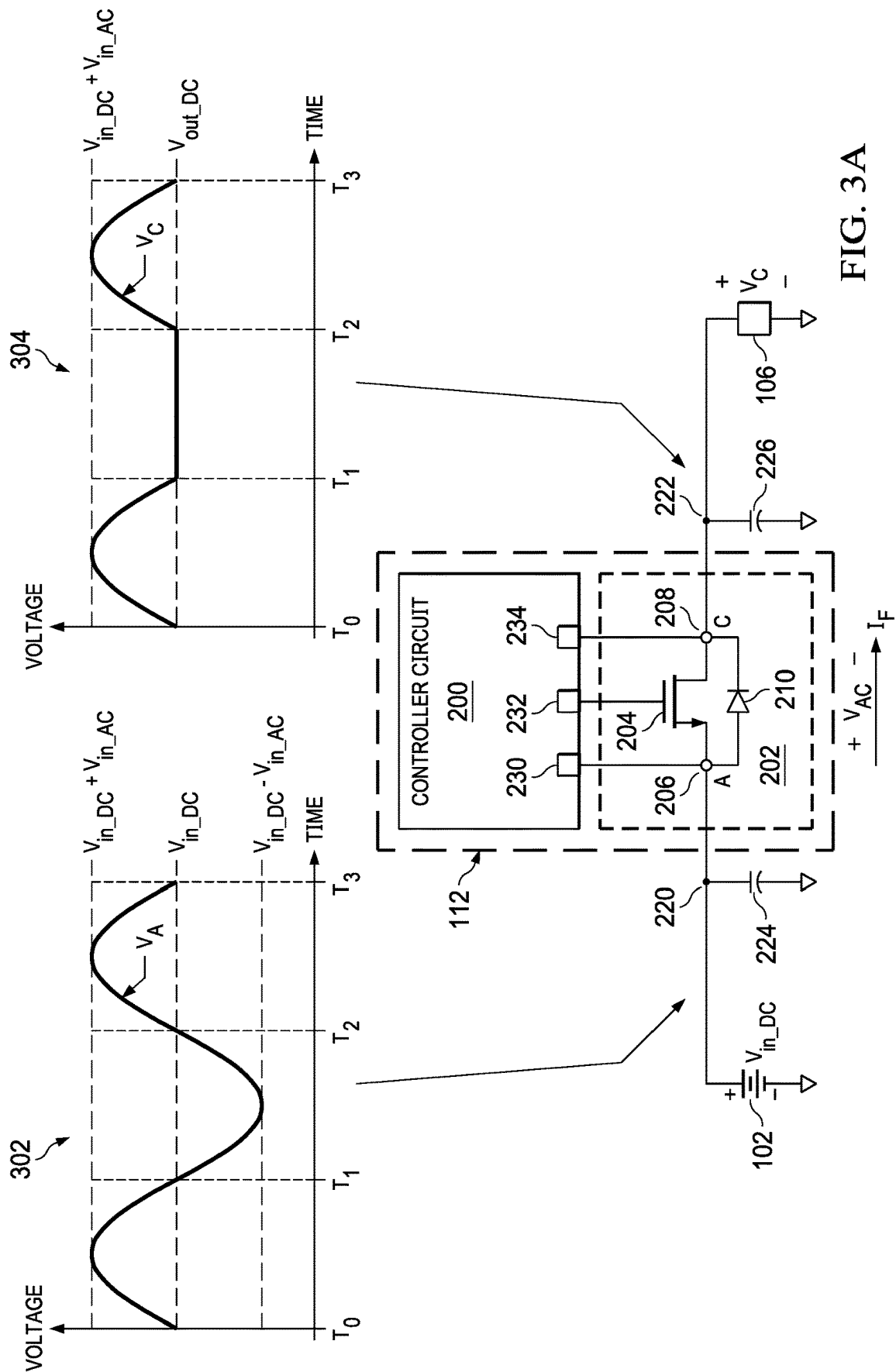


FIG. 3A

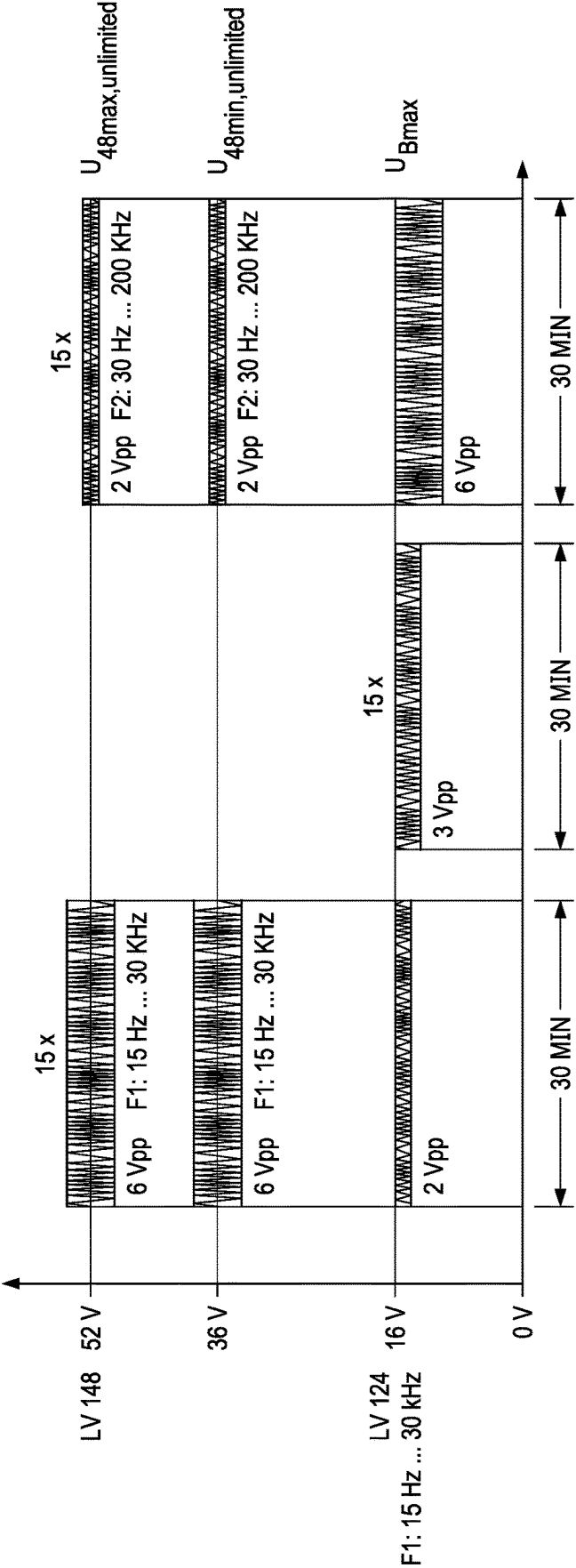
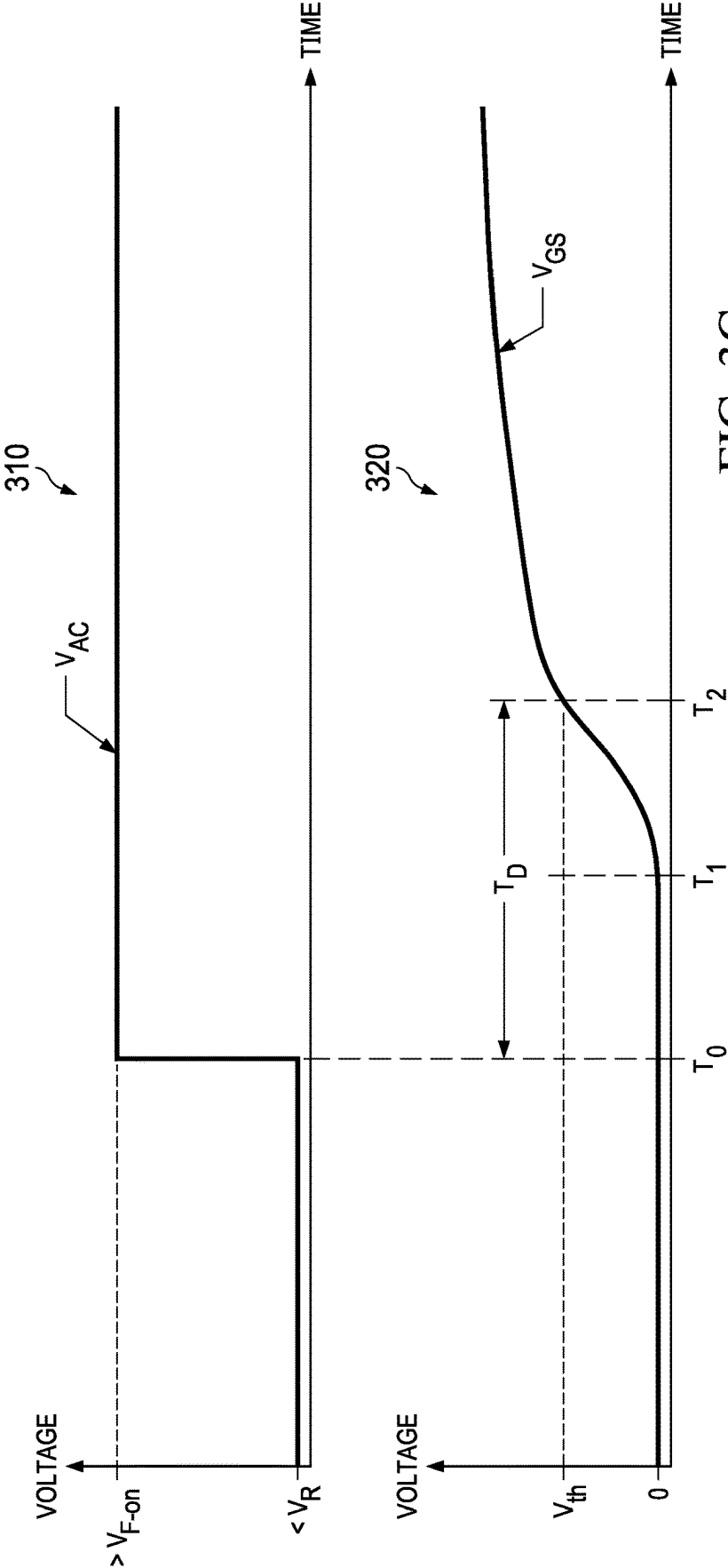


FIG. 3B



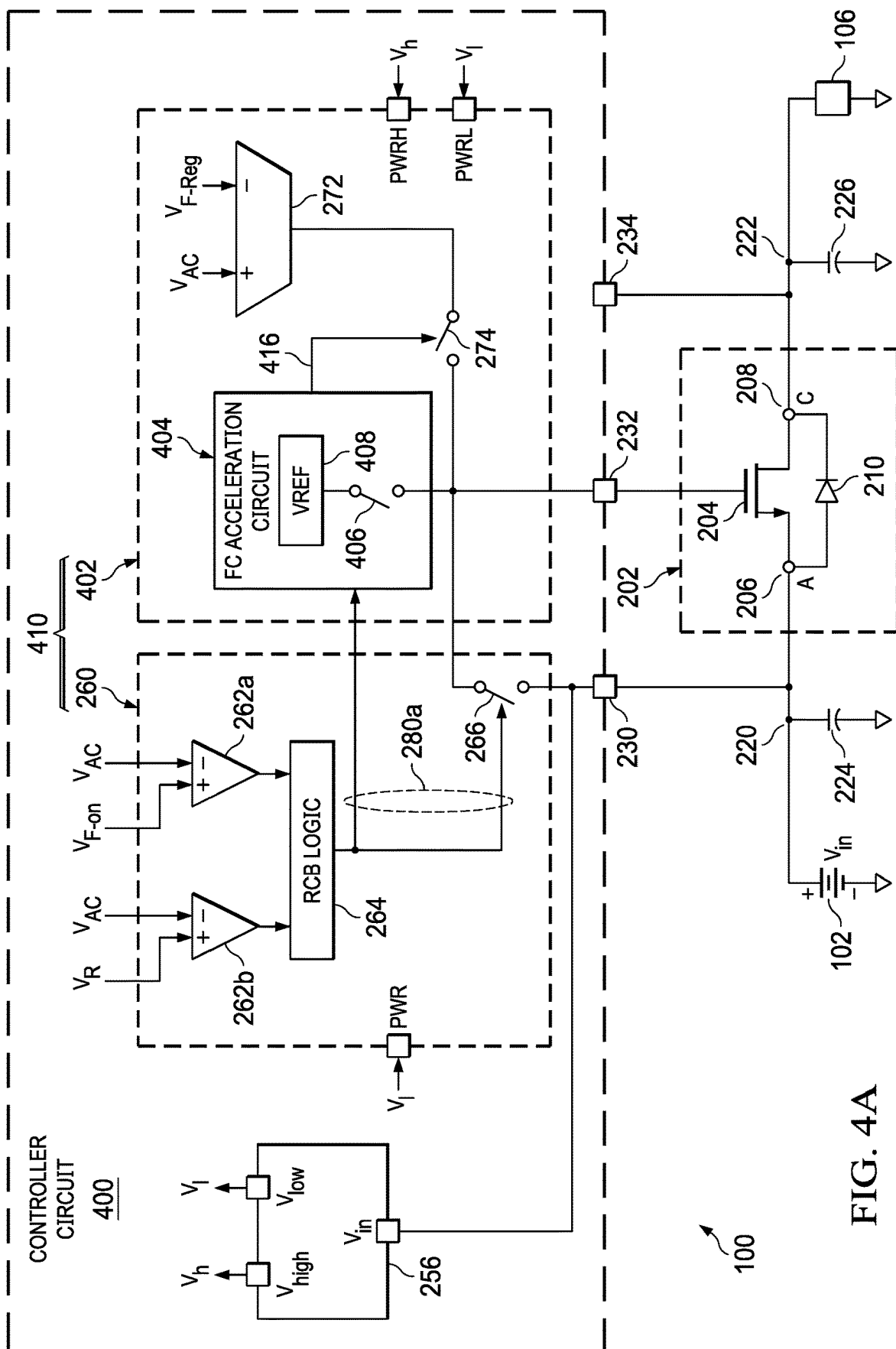


FIG. 4A

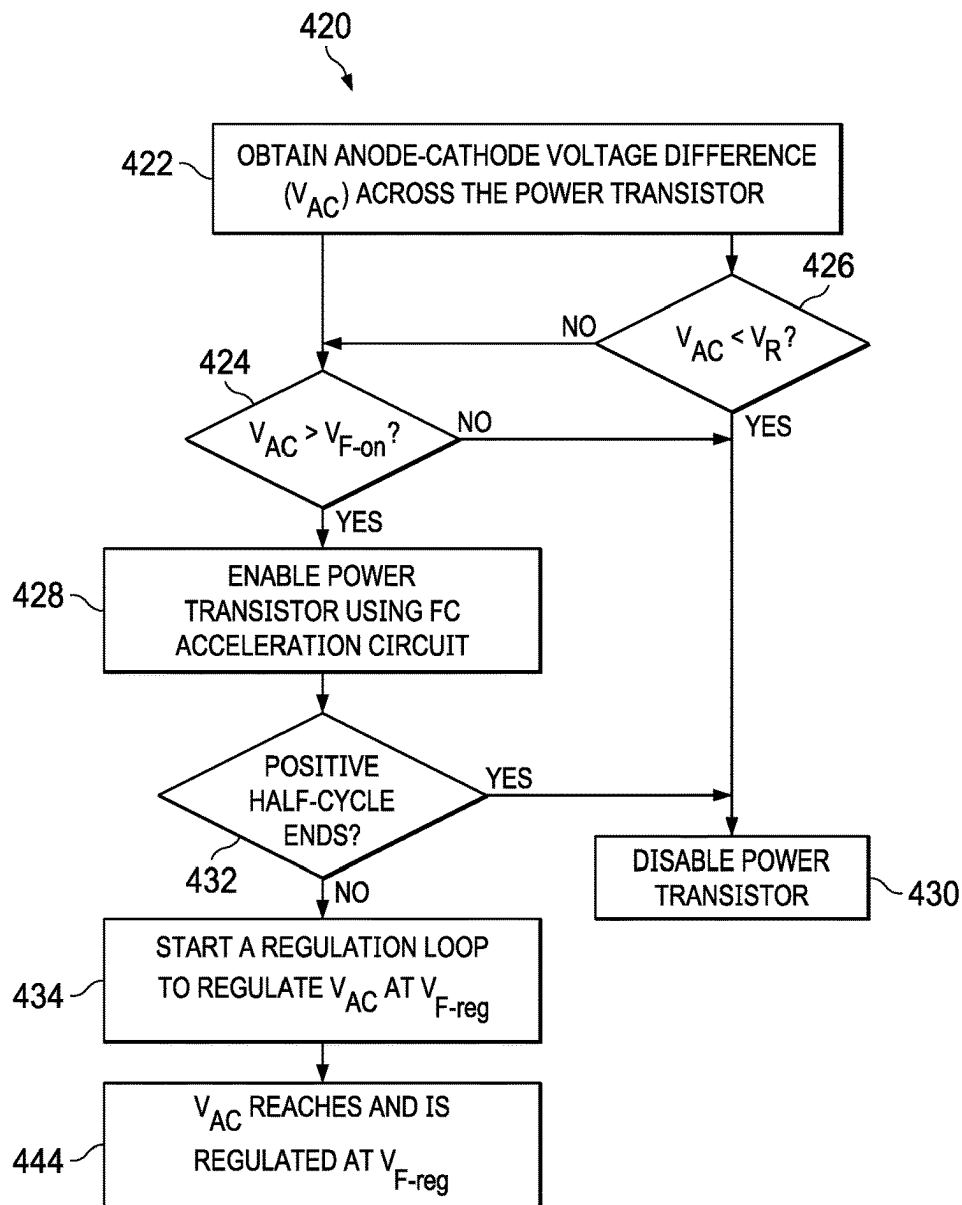


FIG. 4B

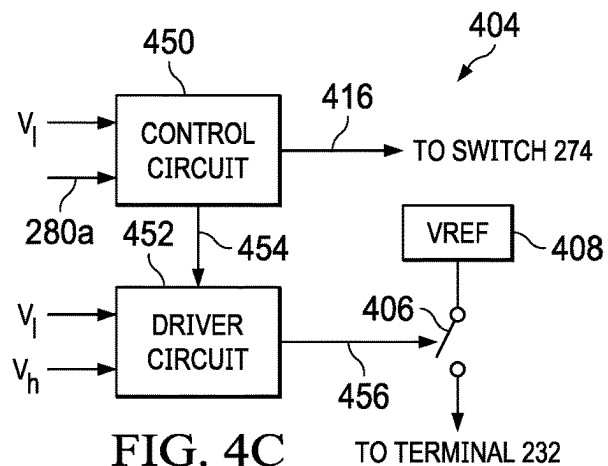
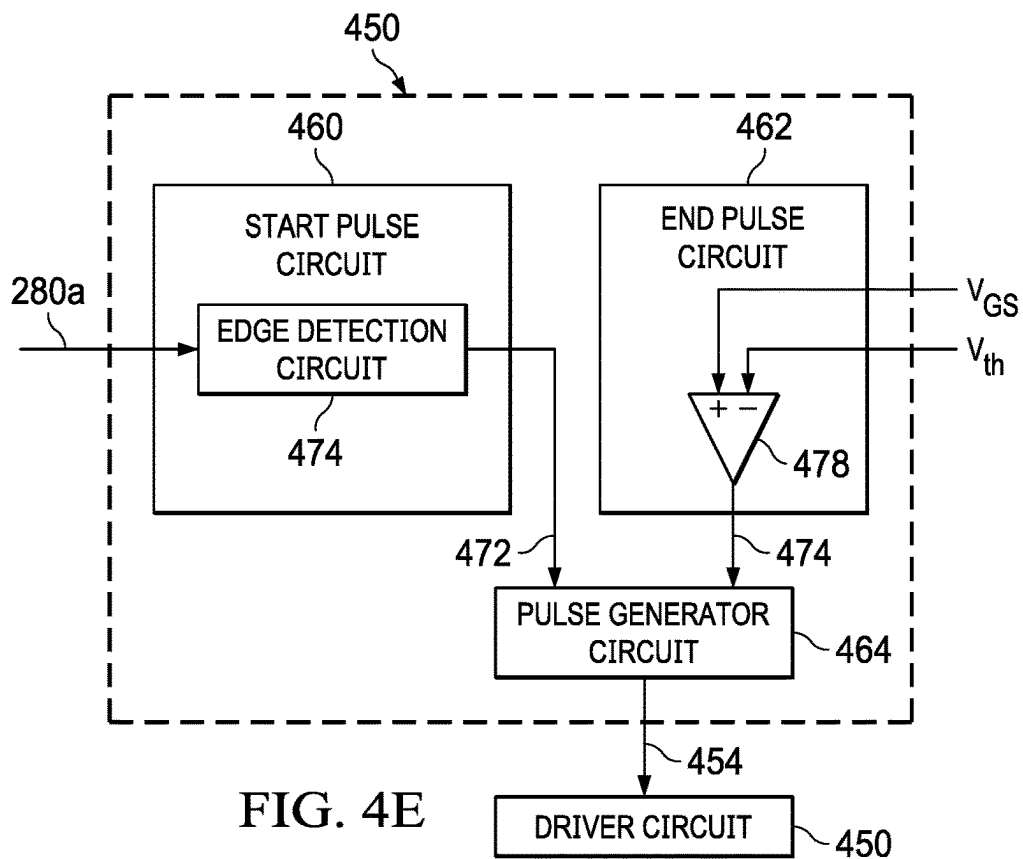
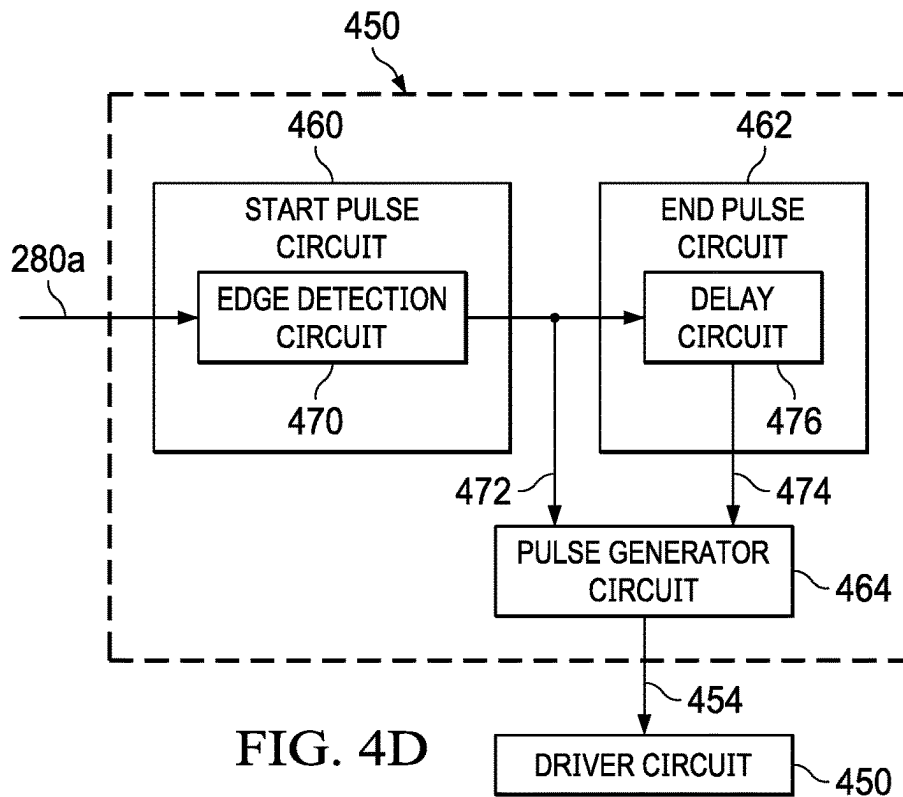


FIG. 4C



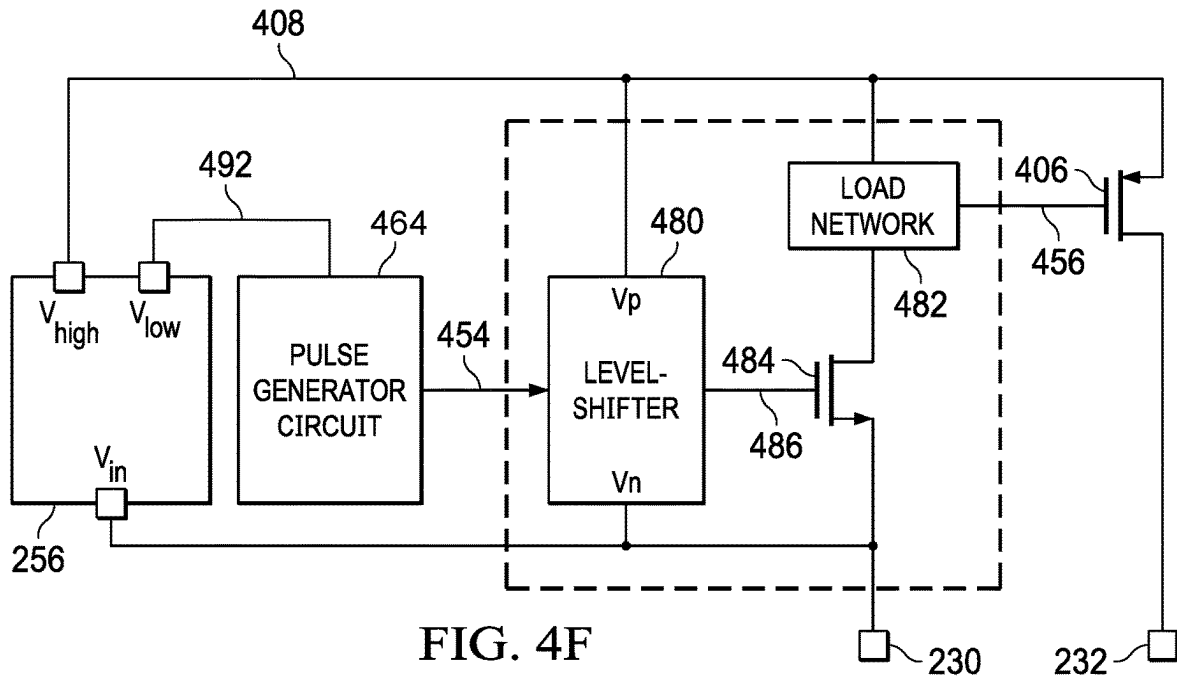


FIG. 4F

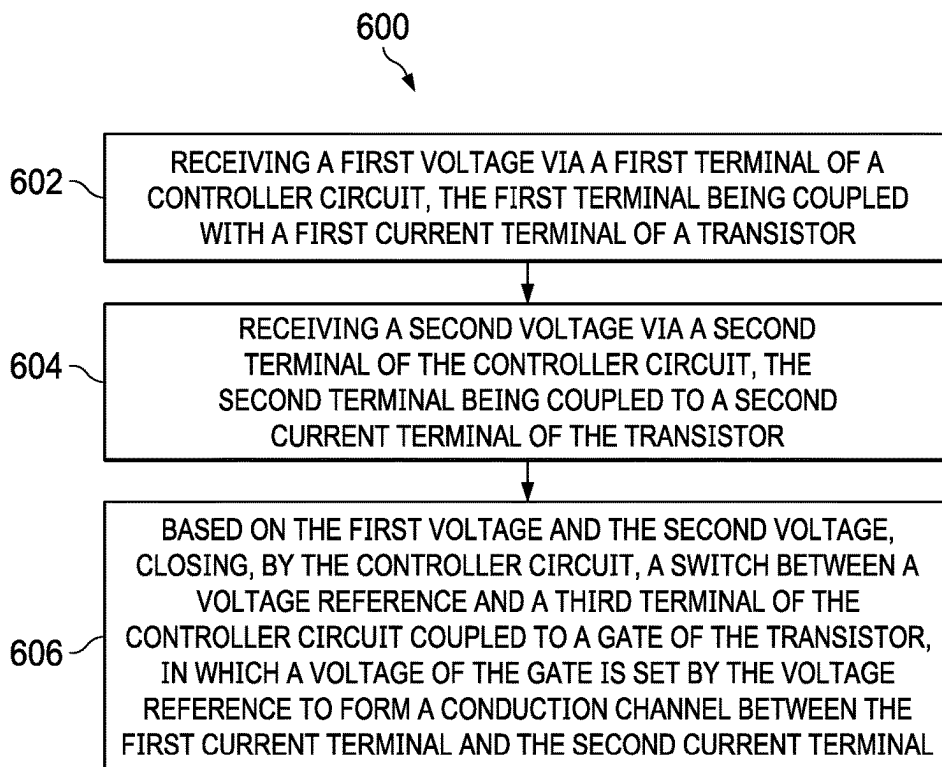
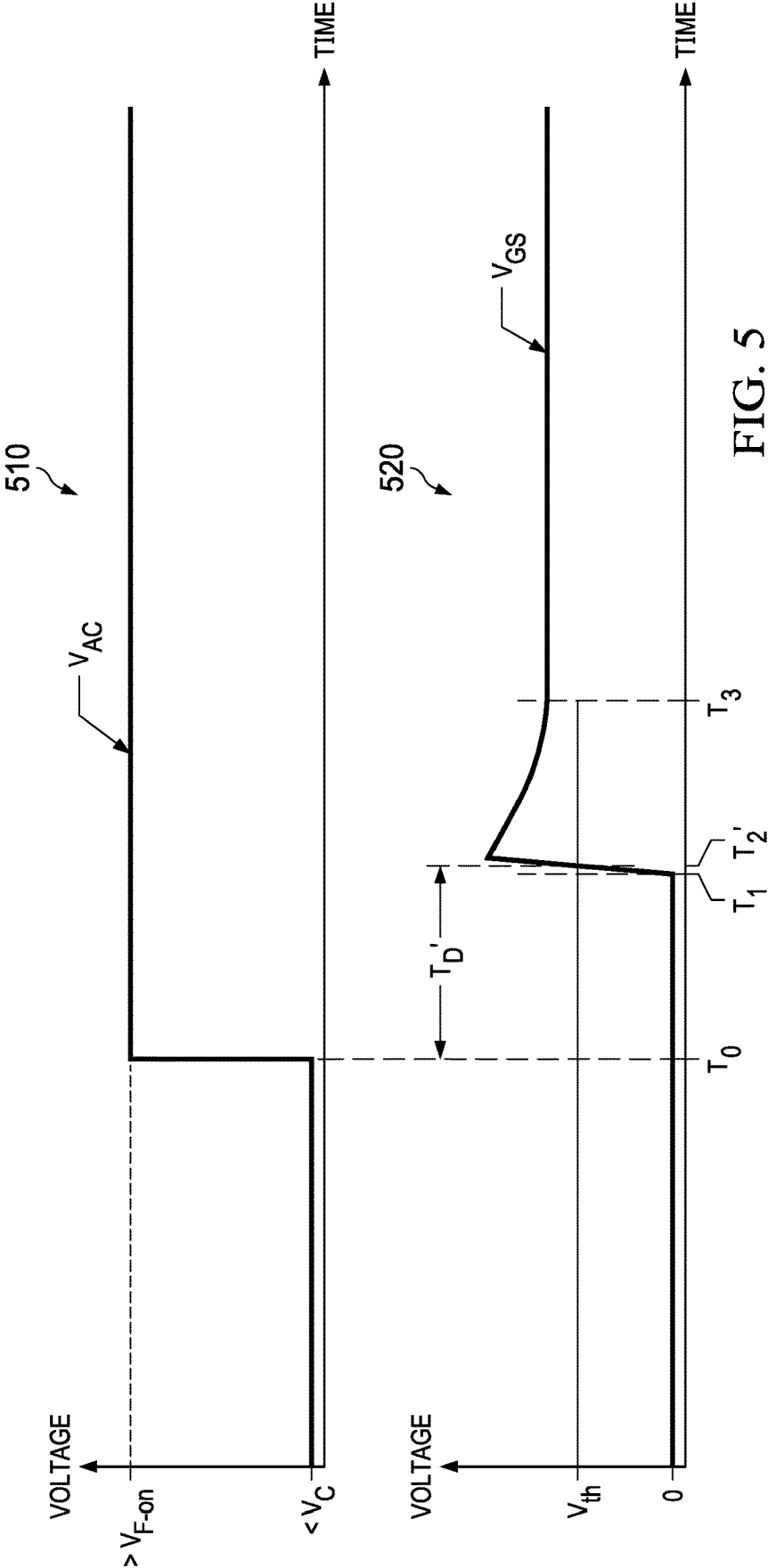


FIG. 6



1

RECTIFICATION BY BATTERY PROTECTION SYSTEM

This application is a continuation of application Ser. No. 17/513,746 filed Oct. 28, 2021, which is incorporated herein by reference.

BACKGROUND

An electric power system that transfers electric power from a battery to a load may include a protection system to protect the load from a reverse battery connection, where the load may receive a negative input voltage from the battery. The protection system can isolate the load from the negative input voltage to prevent the load from being damaged by the negative input voltage. Some examples of the protection system can also block a reverse current from flowing from the load to the battery, to allow the load side additional time to operate before turning off. While the battery may output a direct current (DC) signal, the battery outputs may also include alternating current (AC) signals superimposed on the DC signal. It is also desirable that the protection system can have a short response time, so that the protection system can properly and promptly respond to transient signals output by the battery to protect the load.

SUMMARY

A controller circuit includes a voltage subtractor circuit, an internal voltage generator circuit, and a gate control circuit. The voltage subtractor circuit has a subtractor output and first and second subtractor inputs. The first subtractor input is adapted to be coupled to a first current terminal of a transistor. The second subtractor input is adapted to be coupled to a second current terminal of the transistor. The internal voltage generator circuit has a generator input and a generator output. The generator input is adapted to be coupled to the first current terminal. The gate control circuit has a first gate control input, a second gate control input, and a gate control output. The first gate control input is coupled to the subtractor output. The second gate control input is coupled to the generator output. The gate control output is adapted to be coupled to a gate of the transistor. The gate control circuit also includes a switch coupled between the gate control output and the second gate control input.

In a method, a first voltage is received via a first terminal of a controller circuit, the first terminal being coupled to a first current terminal of a transistor. A second voltage is received via a second terminal of the controller circuit, the second terminal being coupled to a second diffusion of the transistor. Based on the first voltage and the second voltage, a switch between a voltage reference and a third terminal of the controller circuit coupled to a gate of the transistor is closed. A voltage of the gate is set by the voltage reference to form a conduction channel between the first current terminal and the second current terminal.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic diagram of an electric power transfer system in accordance with various examples.

FIGS. 2A through 2F illustrate examples of internal components of a protection system and their operations.

FIGS. 3A, 3B, and 3C illustrate examples of rectification operations performed by the protection system of FIG. 2A-FIG. 2F.

2

FIGS. 4A through 4F illustrate examples of a protection system including internal components to perform a rectification operation in accordance with various examples.

FIG. 5 illustrates examples of voltage graphs depicting a rectification operation by the protection system of FIGS. 4A through 4F.

FIG. 6 is a flowchart of an example method for performing a rectification operation in accordance with various examples.

DETAILED DESCRIPTION

As described above, an electric power system may include a protection system to protect the load from a reverse battery connection. In a case where the electric power system is part of a vehicle, reverse battery connection may occur during maintenance of the vehicle's battery or jump start of the vehicle. Without the protection system, the load may receive a negative voltage from the battery when the battery is reversely connected. The negative voltage can cause huge current to flow from various electronic components of the load, such as electrostatic discharge (ESD) circuits, voltage regulators, etc., which can cause severe damage to these components.

The protection system can include a controller circuit and a transistor. The transistor can include a body diode, of which the anode can be coupled to the battery and the cathode can be coupled to the load. In a reverse battery connection, the battery may output a negative voltage, and the controller circuit can disable the transistor and rely on the reversed-bias body diode to isolate the load from the negative voltage, and to prevent a reverse current from flowing from the load back to the battery. If the battery is connected in the correct polarities, the controller circuit can enable the transistor to transmit a positive voltage and a forward current from the battery to the load. In a case where the anode receives a voltage including both DC and AC components, the protection system can perform a rectification operation to transmit positive AC components (e.g., AC components that adds to the DC component) and not to transmit negative AC components (e.g., AC components that subtracts from the DC component) to the load. For reasons to be described below, the controller circuit may incur substantial delay to enable the transistor, which makes it difficult to perform rectification operations for high frequency AC components. Example techniques described herein speed up the rectification response of the controller circuit, which allows the protection system to perform rectification operations for AC components at a high frequency (e.g., 200 kilo-Hertz (kHz) or above).

FIG. 1 illustrates an example of a system 100. System 100 may include a battery 102, an electric power system 104, and a load 106. Load 106 may include an internal power supply 108, which can include a linear regulator (e.g., a low dropout regulator) and/or a switch mode regulator (e.g., a buck converter, a boost converter, a buck-boost converter, etc.) to provide a supply voltage, and a holdup capacitor to supply a current. Load 106 may further include subsystems 110 that draw power from internal power supply 108. Both internal power supply 108 and subsystems 110 can include various electronic components.

Electric power system 104 is configured to transfer electric power from battery 102 to load 106. Electric power system 104 may receive a voltage V_{in} and a current I_{in} , and provide a voltage V_{out} and a current I_{out} to load 106. Internal power supply 108 can receive voltage V_{out} and current I_{out} from electric power system 104 and provide a voltage

3

$V_{out_internal}$ and a current $I_{out_internal}$ subsystems 110. Voltage V_{out} and current I_{out} provided by electric power system 104 can be based on, respectively, voltage V_{in} and current I_{in} provided by battery 102. Also, voltage $V_{out_internal}$ and current $I_{out_internal}$ can be based on a configuration of internal power supply 108 and subsystems 110. For example, $V_{out_internal}$ can be a fraction of V_{out} to provide a reduced supply voltage required by subsystems 110, and $I_{out_internal}$ can be reduced from I_{in} due to power consumption by electric power system 104 and internal power supply 108. In FIG. 1, the positive terminal of battery 102 can be coupled to electric power system 104 to supply a positive voltage V_{in} , while the negative terminal of battery 102 can be coupled to ground. With such configuration, voltages V_{out} and $V_{out_internal}$ can be positive, and currents I_{in} , I_{out} , and $I_{out_internal}$ can be part of a forward current that flows from battery 102 to load 106.

In some examples, electric power system 104 can include a reverse battery protection system 112 to protect load 106 from a reverse battery connection, where the positive terminal of battery 102 is coupled to ground and the negative terminal of battery 102 is coupled to electric power system 104. As a result, battery 102 may transmit a negative voltage, such as $-V_{in}$, to electric power system 104. Without reverse battery protection system 112, electric power system may transmit the negative voltage to load 106. The negative voltage can cause a huge current to flow from various electronic components of load 106, such as electrostatic discharge (ESD) circuits, voltage regulators of internal power supply 108, etc., which can cause severe damage to these components. Moreover, a reverse current may also flow from load 106 back to battery 102. The reverse current may discharge the holdup capacitor of internal power supply 108 and reduces the holdup capacitor's capability of supplying power to subsystems 110. Reverse battery protection system 112 can isolate load 106 from the negative voltage $-V_{in}$. In some examples, reverse battery protection system 112 can also block the reverse current from flowing from load 106 to battery 102, to allow subsystems 110 additional time to operate before turning off.

FIGS. 2A through 2F illustrate examples of internal components of reverse battery protection system 112 of system 100 and their operations. Referring to the left side of FIG. 2A, reverse battery protection system 112 can include a controller circuit 200 and a transistor 202. Transistor 202 can be an n-channel FET (NFET) or a p-channel FET (PFET). Transistor 202 can have a gate 204, a first current terminal 206, and a second current terminal 208. A body diode 210 can be formed at a p-n junction between first current terminal 206 and second current terminal 208, with first current terminal 206 being an anode (denoted "A" in the figures) and second current terminal 208 being a cathode (denoted "C" in the figures). In a case where transistor 202 is an NFET, first current terminal 206 can be a source whereas second current terminal 208 can be a drain. In a case where transistor 202 is a PFET, first current terminal 206 can be a drain whereas second current terminal 208 can be a source. In system 100, first current terminal 206 can be coupled to battery 102 at a node 220, and second current terminal 208 can be coupled to load 106 at a node 222.

In FIG. 2A, system 100 may include a capacitor 224 and a capacitor 226. Capacitor 224 can model a combination of parasitic capacitances at node 220, such as capacitances of wires and electrical connectors between battery 102 and transistor 202, the junction capacitance at first current terminal 206, etc. Moreover, capacitor 226 can model a combination of parasitic capacitances at node 222, such as

4

capacitances of wires and electrical connectors between load 106 and transistor 202, the junction capacitance at second current terminal 208, etc. Capacitor 226 can also include a physical hold up capacitor to provide a temporary power supply to load 106 when transistor 202 is disabled.

Transistor 202 can be coupled to and controlled by controller circuit 200 to emulate an ideal diode having the same anode and cathode as body diode 210. In some examples, controller circuit 200 can include a terminal 230 adapted to be coupled to first current terminal 206. First current terminal 206 can be the anode of the ideal diode. Controller circuit 200 can also include a terminal 232 adapted to be coupled to gate 204 of transistor 202, and a terminal 234 adapted to be coupled to second current terminal 208 of transistor 202. Second current terminal 208 can be the cathode of the ideal diode. Terminals 230, 232, and 234 can include interconnects (e.g., chip-chip interconnects, traces on printed circuit board (PCB), etc.) that allow signals (e.g., current, voltage, etc.) to flow between controller circuit 200 and transistor 202. Controller circuit 200 can monitor the anode voltage V_A at first current terminal 206 and the cathode voltage V_C at second current terminal 208, and adjust the voltage of gate 204 of transistor 202 via terminal 232 responsive to changes of the anode-cathode voltage V_{AC} to emulate an ideal diode coupled between battery 102 and load 106.

The right side of FIG. 2A illustrates an example transfer function graph 212 of an ideal diode to be emulated by transistor 202. Transfer function graph 212 illustrates a relationship between the amount of a forward current I_F conducted by the diode, from anode to cathode, with respect to a difference voltage between the anode and cathode V_{AC} . If V_{AC} is below a forward voltage V_F , the diode can be reverse-biased, and no forward current (or a minimum amount of forward current) flows through the diode. If V_{AC} is above the forward voltage V_F , the diode is forward bias and can conduct a forward current I_F . When the diode is forward-biased, the anode-cathode voltage V_{AC} can remain constant at V_F independent of the amount of forward current I_F being conducted, so that the cathode voltage V_C can be equal to the anode voltage V_A minus the forward voltage V_F .

To emulate the ideal diode, in a case where V_{AC} is above a forward voltage threshold representing the forward voltage of the ideal diode, controller circuit 200 can increase the gate-source voltage (V_{GS}) of transistor 202 (if transistor 202 is an NFET), or the source-gate voltage (V_{SG}) of transistor 202 (if transistor 202 is a PFET), to be above a threshold voltage V_{th} of the transistor. Raising V_{GS} (V_{SG}) to above V_{th} can turn on/enable transistor 202 by forming a conduction channel between first current terminal 206 and second current terminal 208 under gate 204. The conduction channel can transmit a positive voltage and a forward/positive current from battery 102 to load 106. However, in a case where V_{AC} is below the forward voltage threshold, controller circuit 200 can reduce the gate-source voltage V_{GS} (if transistor 202 is NFET) or source-gate voltage V_{SG} (if transistor 202 is PFET) to be below the threshold voltage V_{th} . Dropping V_{GS} (or V_{SG}) below V_{th} can turn off/disable transistor 202 by removing (or at least reducing) the conduction channel. Body diode 210 is reverse-biased due to V_{AC} being below the forward voltage threshold, and the reverse-biased body diode can block a negative voltage and a reverse/negative current from reaching load 106 from battery 102.

Although transfer function graph 212 shows that an ideal diode has a single forward voltage V_F , in some examples controller circuit 200 can enable a conduction channel of

transistor **202** (between first current terminal **206** and second current terminal **208**) in response to V_{AC} exceeding multiple thresholds, which can indicate that the battery is connected with the correct polarity. Controller circuit **200** can also disable/remove the conduction channel of transistor **202** to block a reverse current/negative voltage in response to V_{AC} being below a reverse bias threshold, which can indicate a reverse battery connection. Such arrangements can improve the robustness of system **100** in light of transient noises.

FIG. 2B illustrates a flowchart of an example method **240** performed by controller circuit **200** in controlling transistor **202**. Method **240** can be performed after controller circuit **200** starts up and has not yet started enabled transistor **202**.

In step **241**, controller circuit **200** can determine an anode-cathode voltage (V_{ac}) across transistor **202**. Controller circuit **200** can monitor the anode voltage (V_A) at terminal **230** and the cathode voltage (V_C) at terminal **234**. Controller circuit **200** can include a subtraction circuit (e.g., implemented using a differential amplifier) to subtract V_C from V_A to obtain V_{AC} .

Controller circuit **200** can then proceed to compare V_{AC} with a forward conduction threshold voltage V_{F-on} , in step **242**. If V_{AC} exceeds V_{F-on} , controller circuit **200** can start a regulation loop to raise the gate-source voltage V_{GS} (or V_{SG} if transistor **202** is PFET) to enable a conduction channel of transistor **202**, and to regulate V_{AC} at a target forward voltage V_{F-reg} , in step **243**. V_{F-reg} can represent V_F of an ideal diode in transfer function graph **212** of FIG. 2A, and transistor **202** can be controlled to emulate a forward-biased diode. In step **244**, V_{AC} reaches (and can be regulated) at V_{F-reg} .

The forward conduction threshold voltage V_{F-on} can be made higher than V_{F-reg} . By having V_{AC} to be higher than V_{F-on} (and to be much higher than V_{F-reg}) to start the forward conduction, the likelihood of mistaking a transient noise at node **220** as a positive voltage supplied by battery **102**, and falsely enabling transistor **202** as a result, can be reduced. The target forward voltage V_{F-reg} can be regulated at a lower voltage than V_{F-on} to reduce voltage drop and power loss across transistor **202** when emulating the forward-biased diode.

Also, controller circuit **200** can compare V_{AC} with a reverse bias threshold voltage V_R , in step **246**, to detect a reverse battery connection. The reverse bias threshold voltage V_R can be a negative voltage that can be received from the negative terminal of battery **102** when the polarity of battery **102** is reversed. Therefore, comparing V_{AC} against a negative voltage to detect a reverse battery connection can reduce the likelihood of false detection of reverse battery connection, such as caused by a transient voltage at node **220**. If V_{AC} is below V_R , which can indicate a reverse battery connection, or if V_{AC} is above V_R but below V_{F-on} , which can indicate a small transient voltage rather than a large positive voltage supplied by battery **102**, controller circuit **200** can maintain transistor **202** in a disabled state, in step **247**. In a case where transistor **202** is disabled and the conduction channel is removed, the reverse-biased body diode **210** can block a negative voltage/a reverse current.

FIG. 2C illustrates examples of internal components of controller circuit **200**. Referring to FIG. 2B, controller circuit **200** can include a gate control circuit **250**. Gate control circuit **250** can include an input **251**, an input **252**, and an output **253**. Input **251** can be adapted to be coupled to first current terminal **206** of transistor **202**, which can be the anode of the diode to be emulated, via terminal **230**. Output **253** can be adapted to be coupled to gate **204** via terminal **232**. Controller circuit **200** further includes a volt-

age subtractor circuit **254**, which can include an op-amp subtractor or other suitable circuits, to receive an anode voltage V_A via terminal **230** and a cathode voltage V_C via terminal **234**, generate an anode-cathode voltage V_{AC} representing a difference between V_A and V_C , and provide V_{AC} to input **252** of gate control circuit **250**. Gate control circuit **250** can generate a gate voltage signal V_G in response to V_{AC} , based on the techniques described in method **240** of FIG. 2B, and provide gate voltage signal V_G via output **253** and terminal **232** to enable transistor **202** to conduct a forward current from the anode to the cathode (and from battery **102** to load **106**), or to disable transistor **202** to block the flow of a reverse current from the cathode back to the anode (and from load **106** back to battery **102**).

Also, controller circuit **200** can include a local voltage generator circuit **256** to generate local voltages. Local voltage generator circuit **256** can receive the anode voltage, which can be a positive voltage provided by battery **102**, via terminal **230** as an input (V_{in}). Local voltage generator circuit **256** can provide a high supply voltage (V_h) to a high power supply terminal (labelled "PWRH" in FIG. 2C) of gate control circuit **250**, and a low supply voltage (V_l) to a low power supply terminal (labelled "PWRL" in FIG. 2C) of gate control circuit **250**. The high supply voltage and the low supply voltage can be generated from the anode voltage V_A and supplied to gate control circuit **250** to reduce the drain-source voltages (V_{DS}) of devices of gate control circuit **250** and the resulting voltage stress. Local voltage generator circuit **256** can include a charge pump to generate the high supply voltage V_h by adding an offset voltage to the anode voltage V_A . Local voltage generator circuit **256** can also include a linear regulator, such as a floating-rail low drop out (LDO) regulator, to generate the low supply voltage V_l by subtracting an offset voltage from the anode voltage V_A .

FIG. 2D illustrates examples of internal components of gate control circuit **250**. Referring to FIG. 2D, gate control circuit **250** can include a reverse current blocking (RCB) circuit **260**, which can include a network of comparators including comparators **262a** and **262b**, an RCB logic circuit **264**, and a switch **266**. Switch **266** is coupled between input **251** (which can be coupled to first current terminal **206**/anode of transistor **202**) and output **253** (which can be coupled to gate **204**). Gate control circuit **250** can also include a forward conduction control circuit **270**, which can include an amplifier **272**, such as an operational transconductance amplifier (OTA), an op-amp, etc., and a switch **274**. Switch **274** can be coupled between the output of amplifier **272** and output **253**. In some examples, switch **274** can be part of a switchable output stage of amplifier **272**. RCB logic **264** can control switches **266** and **274** via a pair of complimentary control signals **280a** and **280b**. Accordingly, when switch **266** is closed, switch **274** can be opened, and vice versa. Gate control circuit **250** can include an inverter **282** to generate control signal **280b** from control signal **280a**.

RCB circuit **260** and forward conduction control circuit **270**, through switches **266** and **274**, can set the gate-source voltage V_{GS} of transistor **202** in response to the anode-cathode voltage V_{AC} , to enable the flow of a forward current from the anode to the cathode (and from battery **102** to load **106**), and to block the flow of a reverse current from the cathode back to the anode (and from load **106** back to battery **102**), based on techniques described in FIG. 2B.

Specifically, referring to FIG. 2E, each of comparators **262a** and **262b** can receive V_{AC} from voltage subtractor circuit **254**. Comparator **262a** can compare V_{AC} against forward conduction threshold voltage V_{F-on} to generate a

first decision, and comparator **262b** can compare V_{AC} against reverse bias threshold voltage V_R to generate a second decision. If V_{AC} is below V_{F-on} (which can indicate V_{AC} is raised by a small transient voltage), or if V_{AC} is below V_R (which can indicate a reverse battery connection), RCB logic **264** can provide a control signal **282a** to disable transistor **202**. Controller circuit **200** can generate control signal **280a** to close switch **266** to connect first current terminal **206** with gate **204**. By connecting first current terminal **206** with gate **204**, the gate voltage V_G can be set to be equal to the source voltage V_S , and the gate-source voltage (V_{GS}) for transistor **202** can be reduced to zero. With the V_{GS} voltage below a threshold voltage V_{th} for forming a channel below gate **204**, transistor **202** can be disabled, and the flow of current between first current terminal **206** and second current terminal **208** of transistor **202** can also be disabled. Moreover, inverter **282** can generate control signal **280b** as a complimentary version of control signal **280a** to open switch **274**, and the output of amplifier **272** can be disconnected from output **253** (and terminal **232**) to avoid interfering with the setting of the gate-source voltage (V_{GS}) for transistor **202** by RCB circuit **260**.

In some examples, RCB logic circuit **264** can include a timing circuit, such as a timer. RCB logic circuit **264** can start the timer after disabling switch **266**. The timer can define an RCB timing window in which transistor **202** is to be continuously disabled regardless of whether V_{AC} is below or above the forward conduction threshold voltage V_{F-on} , and switch **266** is to be continuously enabled. Within the RCB timing window, RCB logic circuit **264** can ignore decisions from comparators **262a** and **262b** to continue closing switch **266** to disable transistor **202**, and continue opening switch **274** to disconnect the output of amplifier **272** from gate **204**. Such arrangements can reduce the likelihood of controller circuit **200** falsely starting a forward conduction due to transient signals at the anode/cathode. The duration of the RCB timing window can be fixed (e.g., built into RCB logic circuit **264**) or can be programmable via a register coupled to RCB logic circuit **264** (not shown in the figures).

FIG. 2F illustrates examples of operations of gate control circuit **250** to enable forward conduction. Referring to FIG. 2F, if the decisions of comparators **262a** and **262b** indicate that V_{AC} is higher than V_{F-on} , which can indicate that the battery is connected in the correct polarities (e.g., positive terminal being coupled to the anode of transistor **202**), RCB logic **264** can generate control signal **280a** to open switch **266**, while control signal **280b**, being a complimentary version of control signal **280a**, can close switch **274** to connect the output of amplifier **272** with output **253** (and gate **204** via terminal **232**). Amplifier **272** is then allowed to adjust the gate voltage V_G (or decrease the gate voltage V_G if transistor **202** is PFET) via output **253** and terminal **232**. With the anode voltage V_A largely fixed by battery **102**, if the gate-source voltage V_{GS} (or V_{SG} if transistor **202** is PFET) becomes higher than the threshold voltage V_{th} of transistor **202**, a conduction channel can be created between first current terminal **206** and second current terminal **208** of transistor **202**. The conduction channel can then enable the flow of forward current I_F from first current terminal **206** to second current terminal **208** of transistor **202** (and from battery **102** to load **106**).

Also, amplifier **272** can implement a feedback loop to set the gate voltage of transistor **202** to regulate the voltage V_{AC} across transistor **202** at a value equal to V_{F-reg} across different forward currents I_F , to emulate a forward-biased diode as shown in FIG. 2A. Amplifier **272** can generate an

output (e.g., a current, a voltage, etc.) that is linearly related to a difference between the anode-cathode voltage V_{AC} and a target forward voltage V_{F-reg} to adjust the on-resistance of the conduction channel of transistor **202**. The current provided by amplifier **272** can be converted to a voltage to set the gate voltage of transistor **202**, which in turn can set the on-resistance of transistor **202**. The on-resistance of transistor **202** can be adjusted, so the voltage V_{AC} across transistor **202** (which can be equal to a product between the on-resistance and the forward current) is maintained at the target forward voltage V_{F-reg} . For example, if load **106** sinks more current, the voltage V_{AC} across transistor **202** can become larger than V_F . In response, amplifier **272** can increase the gate-source voltage V_{GS} of transistor **202** (or the source-gate voltage V_{SG} if transistor **202** is PFET) to reduce the on-resistance of transistor **202**, to reduce the voltage V_{AC} back to V_{F-reg} . However, if load **106** sinks less current, the voltage V_{AC} can decrease. In response, amplifier **272** can reduce the gate-source voltage V_{GS} of transistor **202** (or V_{SG} if transistor **202** is PFET) to increase its on-resistance, to increase the voltage V_{AC} back to V_{F-reg} .

With such arrangements, a voltage V_{AC} across transistor **202** can be maintained to emulate a forward-biased diode. The voltage provided by transistor **202** to load **106** can be maintained constant (or within a narrow range) and can be independent of forward current I_F . This also allows the internal power supply (e.g., internal power supply **108**) of load **106** to provide a stable supply voltage. Moreover, V_{AC} can be maintained at a low value to reduce power loss incurred by transistor **202**, especially when transistor **202** conducts a huge forward current I_F to load **106**.

Referring again to FIG. 2D, RCB circuit **260** can receive low supply voltage V_l from local voltage generator circuit **256**, and forward conduction circuit **270** and inverter **282** can receive both low supply voltage V_l and high supply voltage V_h from local voltage generator circuit **256**. Specifically, RCB circuit **260** can operate within a voltage range below the anode voltage V_A to either disable transistor **202** by shorting the gate and source of transistor **202**, or releasing the gate of transistor **202**, therefore RCB circuit **260** can operate on low supply voltage V_l to reduce voltage stress and to improve reliability of the internal devices of RCB circuit **260**. Moreover, forward conduction control circuit **270** and inverter **282** can operate within a voltage range above the anode voltage V_A . Such arrangements can increase gate overdrive voltage to enable transistor **202** while limiting the gate-drain voltage (V_{GD}) and gate-source voltage (V_{GS}) to reduce voltage stress across transistor **202**, which can improve the reliability of transistor **202**. Also, by operating forward conduction control circuit **270** and inverter **282** between the high supply voltage and the low supply voltage, the voltage swing in the devices of forward control circuit **270** and inverter **282** can be reduced, which can also reduce voltage stress and improve reliability of the internal devices of forward control circuit **270** and inverter **282**.

Referring back to FIG. 1, while battery **102** outputs a DC voltage signal, reverse battery protection system **112** may receive AC voltage signals that are superimposed on the DC voltage output by battery **102**. The AC voltage signals can originate from various sources, such as electromagnetic interference, electrical noises, etc., from other electrical systems. For example, in a case where the system **100** is part of a vehicle, various electrical components, such as motors, switching power converters, etc., can generate periodic AC voltage signals that can be coupled into the wires that couple between battery **102** and electric power system **104**. The AC voltage signals can appear as part of the output voltage of

battery **102** and can be superimposed with the DC voltage signal. The AC voltage signals can have a positive half-cycle and a negative half-cycle. In the positive half-cycle, the AC voltage signals add to the DC voltage signal to provide an increased anode voltage. In a negative half-cycle, in which the AC signals have negative voltages that subtract from the DC voltage signal to provide a reduced anode voltage. Body diode **210** of transistor **202** can switch between a forward-biased condition during the positive half-cycle and a reverse-biased condition periodically during the negative half-cycle. To protect load **106**, controller circuit can control the transistor to operate as a rectifier where the transistor is enabled under the forward-biased condition and is disabled under the reverse-biased condition.

FIG. 3A illustrates an example of a DC voltage signal of battery **102** superimposed with AC voltage signals, and the rectification operation of reverse battery protection system **112**. Referring to the left side of FIG. 3A, graph **302** is an example graph of the anode voltage V_A with respect to time. The output voltage of battery **102** provides a DC input voltage (labelled " V_{in_DC} " in FIG. 3A) to reverse battery protection system **112**. As shown in graph **302**, the anode voltage V_A can include the DC component (V_{in_DC}) as well as an AC component having an amplitude labelled " V_{in_AC} " superimposed with the DC component. The AC component can be attributed to the AC voltage signals. The AC component can be periodic and can have a positive half-cycle in which the AC component has a positive voltage that adds to the DC voltage and a negative half-cycle in which the AC component has a negative voltage that subtracts from the DC voltage. For example, the durations between times T_0 and T_1 and between times T_2 and T_3 can represent positive half-cycles, in which the AC component can add to the DC component, and output voltage of battery **102** (and anode voltage V_A) exceeds the DC component V_{in_DC} and has a maximum voltage of $V_{in_DC} + V_{in_AC}$. Moreover, the duration between times T_1 and T_2 can represent a negative half-cycle, in which the AC components can be subtracted from the DC component, and output voltage of battery **102** (and anode voltage V_A) falls below the DC component V_{in_DC} and has a minimum voltage of $V_{in_DC} - V_{in_AC}$.

Transistor **202** can switch between a forward-biased condition and a reverse-biased condition periodically between each half cycle of the AC signals superimposed with the DC voltage of battery **102**. During the positive half-cycles, the anode voltage (e.g., between V_{in_DC} and $V_{in_DC} + V_{in_AC}$) can be higher than the cathode voltage (e.g., V_{in_DC}) and put transistor **202** in the forward-biased condition, and capacitor **226** can be charged up to V_{in_DC} . During the negative half-cycles, the anode voltage (e.g., between $V_{in_DC} - V_{in_AC}$ and V_{in_DC}) can be lower than the cathode voltage (e.g., V_{in_DC}) and put transistor **202** in the reverse-biased condition. As part of the rectification operation to prevent a reverse current from flowing back from load **106** to battery **102** during the negative half-cycles, controller circuit **200** can enable transistor **202** during the positive half-cycles and disable transistor **202** during the negative half-cycles.

Graph **304** of FIG. 3A shows an example graph of the cathode voltage V_C with respect to time as a result of a rectification operation performed by controller circuit **200** with transistor **202**. Specifically, during a positive half-cycle, such as between times T_0 and T_1 , the anode-cathode voltage V_{AC} can exceed the forward conduction threshold voltage V_{F-on} , and controller circuit **200** can enable transistor **202** to form a conduction channel between first current terminal **206** and second current terminal **208** to connect battery **102** to load **106**. Accordingly, the cathode voltage V_C

across load **106** can track the anode voltage V_A . The voltage V_{AC} can also be maintained at the target forward voltage V_{F-reg} by amplifier **272**. Accordingly, the cathode voltage V_C can have a DC component of V_{out_DC} which equals $V_{in_DC} - V_{F-reg}$, as well as an AC component having the same (or similar) amplitude V_{in_AC} as the AC signals at the anode. Within the positive half-cycle, the cathode voltage V_C can first increase from V_{out_DC} , reaching a peak of $V_{out_DC} + V_{in_AC}$, and then decrease back to V_{out_DC} .

Moreover, during a negative half-cycle (e.g., between times T_1 and T_2), the anode voltage V_A is reduced by the AC component, while the cathode voltage V_C can be held at V_{out_DC} by holdup capacitor **226**, and the anode-cathode voltage V_{AC} can be below the reverse bias threshold voltage V_R . This can cause controller circuit **200** to disable transistor **202** by removing/disabling the conduction channel between first current terminal **206** and second current terminal. Accordingly, the cathode voltage V_C can stop tracking the anode voltage V_A during the negative half-cycle. The negative half-cycle is followed by a subsequent positive half-cycle (e.g., between times T_2 and T_3), in which controller circuit **200** can enable transistor **202** again to allow a forward conduction from battery **102** to load **106**, and the cathode voltage V_C can track the anode voltage V_A again.

The AC voltage signals that superimpose with the DC voltage signal output by battery **102** can have a high frequency, which can lead to high frequency changes in the input voltage to protection system **112**. FIG. 3B illustrates examples of AC voltage signals that may be received by protection system **112** in a vehicle application. For example, according to LV **124** and LV **148**, which define testing of electronic components for vehicles, the electric systems of a vehicle can have AC voltage signals of 2-6 peak-to-peak voltage (V_{pp}) at a frequency of 15 Hz to 200 kHz superimposed on the DC voltage signal output by a battery. Assuming the AC voltage signals have a frequency of 200 kHz and a cycle time of 5 micro-seconds (us), to perform rectification, controller circuit **200** may need to be able to repeatedly enable transistor **202** within a positive half-cycle of 2.5 us and disable transistor **202** within a negative half-cycle of 2.5 us. Accordingly, it is desirable that the controller circuit **200** can have a short rectification response time, in order to enable and disable the transistor promptly in response to high-frequency AC voltage signals present at the battery output.

But controller circuit **200** may have a long rectification response time, especially in enabling transistor **202** to start forward conduction, which makes it challenging to handle AC ripples up at a frequency of 200 kHz and beyond. Specifically, as described above, to emulate a forward-biased diode having a constant forward voltage, controller circuit **200** (and forward conduction control circuit **270**) may include amplifier **272**, which can be linear amplifier such as an OTA, to implement a feedback loop to regulate the anode-cathode voltage V_{AC} across the transistor at the target forward voltage V_{F-reg} . The output of the amplifier can be linearly related to V_{AC} . But the amplifier may have a low gain (e.g., a low transconductance (gm) for OTA) to improve loop stability, especially for low forward current/load current. The low gain can reduce the speed by which the amplifier can raise the gate voltage of the transistor, which in turn increases the time it takes for controller circuit **200** to enable transistor **202** (by forming a conduction channel between first current terminal **206** and second current terminal **208**) in the positive half-cycle of an AC ripple.

FIG. 3C illustrates a graph **310** and a graph **320**. Graph **310** provides an example of variations of V_{AC} with respect

to time, and graph 320 provides an example of variations of gate-source voltage V_{GS} of transistor 202 with respect to time, where transistor 202 is under the control of controller circuit 200. Referring to graph 310, V_{AC} can increase from a first voltage below V_R to a second voltage above V_{F-on} at time T_0 . In response to detecting that V_{AC} increases from below V_R to above V_{F-on} , gate control circuit 250 to switch from disabling transistor 202 to enabling transistor 202 by changing the gate voltage.

Referring to graph 320, the gate-source voltage V_{GS} is initially at 0V and transistor 202 is disabled prior to time T_0 . In response to detecting the transition of V_{AC} at time T_0 , gate control circuit 250 may maintain V_{GS} at 0V, and transistor 202 can remain in the disabled state, until time T_1 . At time T_1 , gate control circuit 250 can start increasing V_{GS} . At time T_2 , V_{GS} reaches the threshold voltage V_{th} of transistor 202. In response to V_{GS} reaching V_{th} , transistor 202 can form a conduction channel between first current terminal 206 and second terminal 208, and transistor 202 is enabled. Accordingly, a total delay T_D has elapsed from the time T_0 when the anode-cathode voltage V_{AC} transitions from lower than V_R to higher than V_{F-on} to the time T_2 when transistor 202 is enabled.

The delay T_D can be attributed to various sources. For example, a first part of the delay T_D , between time T_0 and T_1 , can be attributed to the delay incurred by RCB 260 in detecting the changes in V_{AC} and opening switch 266 (and closing switch 274) to allow amplifier 272 to start increasing gate voltage V_G . Moreover, a second part of the delay T_D , between times T_1 and T_2 , can be incurred by amplifier 272 in raising the gate voltage V_G as part of the feedback loop to regulate V_{AC} at V_{F-reg} . The rate at which amplifier 272 increases V_G can be based on, for example, the gain (e.g., transconductance) of amplifier 272, the capacitance of gate 204, etc.

The total delay T_D between time T_0 and time T_2 can represent a rectification response time of protection system 112. Depending on various factors, such as comparator delay, the transconductance of amplifier 272, the capacitance of gate 204, etc., the rectification response time T_D can be near 2 us. In a case where the AC ripples have a frequency of 200 kHz and beyond, the half-cycle period will be less than 2.5 us. With a rectification response time that spans most of the positive half-cycle period, transistor 202 can remain disabled for most of the positive half-cycle period, which can prevent load 106 from receiving electric power from battery 102 during most of the positive half-cycle period. Accordingly, protection system 112 may be unable to perform the rectification operation in response to high-frequency AC signals present at the output of battery 102.

FIG. 4A through FIG. 4F illustrate examples of a controller circuit 400 that can address at least some of the issues. As shown in FIG. 4A, controller circuit 400 can be part of system 100 of FIG. 1 and can include RCB circuit 260 of FIGS. 2A through 2F. In addition, controller circuit 400 can include a forward conduction (FC) control circuit 402, which can include an FC acceleration circuit 404, as well as amplifier 272 and switch 274 as described in FIGS. 2A through 2F. RCB circuit 260 and FC control circuit 402 can be part of a gate control circuit 410, and each of RCB circuit 260 and FC control circuit 402 is coupled with terminal 232 to set the gate voltage of transistor 202.

FC control circuit 402 can set a gate voltage to form a conduction channel between first terminal 206 and second current terminal 208, to connect battery 102 to load 106. FC acceleration circuit 404 can first set the gate voltage to form the conduction channel, followed by amplifier 272 adjusting

the gate voltage to regulate the V_{AC} voltage. Specifically, FC acceleration circuit 404 can include a switch 406 coupled between terminal 232 (and gate 204 of transistor 202) and a voltage reference 408 (labelled "VREF" in FIG. 4A). Switch 406 can be an NFET, a PFET, or a parallel combination of both. FC acceleration circuit 404 can receive an indication of a forward-biased condition. The indication can be based on control signal 280a. For example, based on detecting a transition of control signal 280a to a state to open switch 266, FC acceleration circuit 404 can determine that the forward-biased condition is present (with V_{AC} above V_{F-on}), and close switch 406. The closing of switch 406 can connect voltage reference 408 with gate 204 of transistor 202 via terminal 232, which can raise the gate voltage (or reduce the gate voltage if transistor 202 is PFET) to enable transistor 202. Voltage reference 408 can provide a target voltage to enable transistor 202. After transistor 202 is enabled, if protection system 112 is still within the positive half-cycle, forward conduction circuit 402 can open switch 406, and provide a control signal 416 to close switch 274, which connects the output of amplifier 272 to terminal 232 and allows amplifier 272 to start a regulation loop to further adjust the gate voltage.

Components of FC control circuit 402, including amplifier 472 and FC acceleration circuit 404, can operate on a high supply voltage V_h and a low supply voltage V_l generated by local voltage generator circuit 256 from anode voltage V_A . For example, voltage reference 408 can be provided by the high supply voltage V_h if transistor 202 is an NFET. Voltage reference 408 can also be provided by low supply voltage V_l if transistor 202 is a PFET. The high supply voltage V_h and low supply voltage V_l can be configured based on, for example, a margin above (or below) a threshold voltage V_{th} of transistor 202 for forming the conduction channel to enable the transistor. In some examples, the high supply voltage V_h and low supply voltage V_l can also be configured based on a voltage stress threshold for the V_{GS}/V_{SG} voltage, to improve reliability of transistor 202. Further, switch 406 can be controlled by a control signal that swings between V_h and V_l to reduce the on-resistance as well as the voltage stress on switch 406, which can improve both the bandwidth and reliability of switch 406.

With the arrangements of FIG. 4A, forward conduction control circuit 402 need not rely on amplifier 272 to both start the enabling of transistor 202 and regulate the V_{AC} of transistor 202, and the reduced gain/transconductance of amplifier 272 (e.g., due to the constraint of loop stability) can have less impact on the delay in enabling transistor 202. Moreover, the speed of enabling transistor 202 can be further improved by increasing the bandwidth (e.g., by reducing the on-resistance) of switch 406. All these can reduce the rectification response time of controller circuit 400 and improve the handling of high-frequency AC signals at the output of battery 102.

FIG. 4B illustrates a flowchart of an example method 420 performed by controller circuit 400 in controlling transistor 202. Method 400 can be performed after controller circuit 400 starts up and has not yet enabled transistor 202, and with battery 102 supplying a DC voltage to transistor 202. AC ripples may be coupled into and superimposed with the DC voltage, and appear as an AC component of the anode voltage V_A of transistor 202.

In step 422, controller circuit 400 can determine an anode-cathode voltage (V_{AC}) across transistor 202. Controller circuit 400 can monitor the anode voltage (V_A) at terminal 230 and the cathode voltage (V_C) at terminal 234.

Controller circuit 400 can include a subtraction circuit (e.g., implemented using a differential amplifier) to subtract V_C from V_A to provide V_{AC} .

Controller circuit 200 can compare V_{AC} with forward conduction threshold voltage V_{F-on} using comparator 262a, in step 424. In addition, controller circuit 200 can also compare V_{AC} with reverse conduction threshold voltage V_R , in step 426. V_{AC} can exceed V_{F-on} (and V_R) during a positive half-cycle of the AC signals/ripples, and can be below V_{F-on} or V_R during a negative half-cycle of the AC signals/ripples, or when the battery is reverse connected.

In step 428, if V_{AC} exceeds V_{F-on} , RCB blocking logic 264 can output control signal 280a to open switch 266 to disconnect RCB circuit 260 from the gate of transistor 202. The opening of switch 266 can also enable FC control circuit 402 to set the gate voltage of transistor 202. Moreover, based on control signal 280a is in a state to open switch 266, FC acceleration circuit 404 can close switch 406 to raise the gate voltage of transistor 202 (or reduce the gate voltage if transistor 202 is PFET). As to be described below, FC acceleration circuit 404 can close switch 406 for a pre-determined duration, or until the gate-source voltage exceeds the threshold voltage V_{th} of transistor 202 (or falls below V_{th} for PFET). Moreover, if V_{AC} is below V_{F-on} or V_R (in steps 424 and 426), RCB logic 264 can close switch 266 to bring the gate-source voltage V_{GS} of transistor 202 to zero to disable the transistor, and switch 406 can be open, in step 430.

Referring back to step 428, after transistor 202 is enabled by FC acceleration circuit 404 and a conduction channel is formed between first current terminal 206 and second current terminal 208, controller circuit 400 can determine whether the positive half-cycle ends, in step 432. The determination can be based on comparing a new V_{AC} (obtained after transistor 202 is enabled) against V_{F-on} and V_R . If the new V_{AC} is below either V_{F-on} or V_R , controller circuit 400 can determine that the positive half-cycle has ended, and can proceed to step 430 to disable transistor 202.

Moreover, if the new V_{AC} is above V_{F-on} in step 432, which can indicate that the positive half-cycle has not yet ended, controller circuit 400 can start a regulation loop to regulate V_{AC} at a target forward voltage V_{F-reg} , in step 434. As part of step 434, FC acceleration circuit 404 can transmit control signal 416 to close switch 274, which allows amplifier 272 to set the gate voltage of transistor 202 via terminal 232. Amplifier 272 can also implement a feedback loop in which the output of amplifier 272 is linearly related to a difference between V_{AC} and V_{F-reg} , as described above. In step 444, V_{AC} reaches V_{F-reg} , and can be regulated at V_{F-reg} by amplifier 272.

FIGS. 4C through 4F illustrate examples of additional internal components of FC acceleration circuit 404. Referring to FIG. 4C, in addition to switch 406 and voltage reference 408, FC acceleration circuit 404 can include a control circuit 450 and a driver circuit 452. Control circuit 450 can receive control signal 280a. In response to receiving control signal 280a, control circuit 450 can provide control signal 416 to switch 274, and a control signal 454 to switch 406. Control circuit 450 can operate with the low supply voltage V_L . Driver circuit 452 can receive control signal 454 and convert it to a control signal 456 that swings between a pre-determined minimum voltage v_{min} and high supply voltage V_H , and provide control signal 456 to switch 406.

FIG. 4D and FIG. 4E illustrate examples of internal components of control circuit 450. Control circuit 450 can include a start pulse circuit 460, an end pulse circuit 462, and a pulse generator 464. Pulse generator 464 can generate

control signal 454, which can be in the form a pulse signal, to close switch 406 for a finite duration, with the start time and end time of the pulse signal being defined by, respectively, start pulse circuit 460 and end pulse circuit 462. Start pulse circuit 460 can include an edge detection circuit 470 to detect a transition in control signal 280a that indicates the start of a forward-biased condition. For example, the transition of control signal 280a can be to a state to open switch 266. Responsive to detecting the transition, start pulse circuit 460 can provide a start signal 472 that reflects the timing of the transition to pulse generator circuit 464. In response to receiving start signal 472, pulse generator circuit 464 can start the pulse of control signal 454.

In addition, end pulse circuit 462 can provide an end signal 474 to pulse generator circuit 464 to end the closing of switch 406. End pulse circuit 462 can generate end signal 474 based on various techniques. Referring to FIG. 4D, end pulse circuit 462 can include a delay circuit 476 to provide end signal 474 as a delayed version of start signal 472, with the delay setting a width of the pulse of control signal 454. Delay circuit 476 can include delay elements (e.g., buffers) to delay pull-up start signal 470. In some examples, delay circuit 476 can include programmable delay elements such that the delay introduced to start signal 472 in generating end signal 474 is programmable. In such examples, delay circuit 476 can be coupled with a programming register to receive a setting for the delay.

The delay of delay circuit 476 (and the pulse width) can be set based on various criteria. For example, the delay can be set based on the minimum half-cycle period of the AC ripples, such that the enabling of the pull-up path can start and end within a positive half-cycle period for the highest-frequency AC ripples to be rectified by controller circuit 400. The delay can also be set based on other factors, such as the gate capacitance of transistor 202, which can vary based on the dimension of the transistor. In some examples, the delay can also be set based on a target on-resistance of transistor 202 for the forward conduction operation. For example, based on the threshold voltage of transistor V_{th} , a target gate voltage of transistor 202 can be determined, which can be a certain percentage (e.g., 90%) of the voltage provided by voltage reference 408. Based on the gate capacitance of transistor 202 as well as the on-resistance of switch 406, the time need to bring the gate voltage of transistor 202 to the target voltage can be determined, and the pulse width (and delay of delay circuit 476) can be pre-configured accordingly.

FIG. 4E illustrates another example of pulse end circuit 462. Specifically, pulse end circuit 462 can end the pulse and open switch 406 when the gate-source voltage V_{GS} goes above a threshold, such as threshold V_{th} of transistor 202 for forming a conduction channel between first current terminal 206 and second current terminal 208. Referring to FIG. 4E, pulse end circuit 462 can include a comparator 478 to compare the gate-source voltage V_{GS} (or V_{SG} for PFET) which can be obtained from a subtraction circuit that subtracts the source voltage V_S (obtained via terminal 230) from the gate voltage V_G (obtained via terminal 232). If V_{GS} rises above V_{th} , comparator 478 can trip to generate pull-up end signal 472, which can signal the end of the pull-up operation to pulse generator 466.

FIG. 4F illustrates examples of internal components of driver circuit 450, as well as switch 406 and voltage reference 408, in a case where transistor 202 is an NFET. Referring to FIG. 4F, switch 406 can be implemented using a PFET, coupled between voltage reference 408 and terminal 232. Voltage reference 408 can be provided as the high

15

supply voltage V_h from V_{high} terminal of local voltage generator circuit 256. In addition, driver circuit 450 includes a level-shifter 480, a load network 482, and an NFET 484 coupled between load network 482 and terminal 230. Level-shifter 480 can have a positive voltage supply terminal (labelled " V_p ") coupled with high supply voltage V_h (and voltage reference 408), and a negative voltage supply terminal (labelled " V_n ") coupled with terminal 230, which provides access to the anode voltage V_A . Level-shifter 480 can receive, from pulse generator circuit 464, control signal 454 that can swing between 0 to V_h and perform a level-shifting of control signal 454 to generate a control signal 486 with a voltage swing between V_h and V_A , and drive NFET 484 with control signal 486. Both control signals 452 and 486 can be pulse signals.

In the example of FIG. 4F, control signals 454 and 486 can be active-high pulse signals, with the high pulse defining the duration when switch 406 is closed. When NFET 484 is enabled by control signal 486, NFET 484 can form a voltage divider with load network 482, and the voltage divider can set the minimum voltage V_{min} of control signal 456 being at V_{min} . PFET/switch 406 can be enabled/closed to connect gate 204 of transistor 202 to voltage reference 408. When NFET 484 is disabled by control signal 486, load network 482 can pull control signal 456 to the high supply voltage V_h , and PFET/switch 406 can be disabled/opened. Load network 482 can include a network of passive elements, such as resistors, Zener diodes, etc., to limit the voltage swing of the gate of pull-up transistor 480 to reduce voltage stress and to improve reliability.

With the arrangements of FIG. 4F, the gate voltage V_G can be set at a value higher than the anode voltage V_A (e.g., high supply voltage V_h) to enable transistor 202. As the anode voltage V_A is also the source voltage V_S of transistor 202, the gate-source voltage V_{GS} can exceed the threshold voltage V_{th} to form the conduction channel between first current terminal 206 and second current terminal 208 of transistor 202. Moreover, by using level-shifter 480 to level-shift a pulse signal that swings between low supply voltage V_l and ground to another pulse signal that swings between high supply voltage V_h and the anode voltage V_A , the maximum voltages experienced by the transistor devices of pulse generator circuit 464, as well as NFET 484 and PFET/switch 406, can be reduced, which can reduce voltage stress and improve reliability of control circuit 400. Further, as the anode voltage V_A (and source voltage V_S) may vary based on the DC voltage output by battery 102, generating the control signals and internal voltages based on the anode voltage V_A can ensure that the control signals and internal voltages have a deterministic relationship with the anode voltage V_A and source voltage V_S , which can improve the predictability of control circuit 400.

FIG. 5 illustrates examples of a graph 510 of V_{AC} across transistor 202 with respect to time, and a graph 520 of the gate-source voltage V_{GS} of transistor 202 with respect to time, under the control of controller circuit 400. Referring to graph 510, V_{AC} increases from a first voltage below V_R to a second voltage above V_{F-on} at time T_0 . Referring to graph 520, the gate-source voltage V_{GS} is initially at 0V and transistor 202 is disabled. It takes until time T_1 when RCB circuit 260 generates control signal 280a to disable switch 266.

Between times T_1 and T_2' , responsive to detecting that control signal 280a is at a state to open switch 266, FC acceleration circuit 404 can close switch 406 to raise (or reduce) the gate voltage of transistor 202. The rate of change

16

of the gate voltage can be impacted more by, for example, the on-resistance of switch 406, the capacitance of gate 204, etc., and less by the gain/transconductance of amplifier 472. Accordingly, in FIG. 5, the V_{GS} voltage can change at a higher rate and can reach the threshold V_{th} at time T_2' , which is earlier than time T_2 in FIG. 3C. Accordingly, the total delay T_D' between times T_1 and T_2' can also be reduced compared with T_D in FIG. 3C. Switch 406 can be opened at time T_2' . Between times T_2' and T_3 , amplifier 272 can operate a regulation loop to set the V_{GS} voltage in order to regulate the anode-cathode voltage V_{AC} at V_{F-REG} .

FIG. 6 is a flowchart of an example method 600 for performing a rectification operation by an electric power system coupled between a battery and a load, in a case where the DC voltage signal output by the battery is superimposed with AC signals. Method 600 can be performed by, for example, controller circuit 400 of FIGS. 4A through 4F. Although the method steps are described in conjunction with FIGS. 4A through 4F, any system configured to perform the method steps, in any suitable order, is within the scope of this description.

At step 602, controller circuit 400 can receive, via a first terminal (e.g., terminal 230), a first voltage. The first terminal can be adapted to be coupled to a transistor's first current terminal (e.g., transistor 202's first current terminal 206). The first current terminal can be an anode of a diode. The first current terminal can be a source if the transistor is NFET, and can be a drain if the transistor is PFET. The first current terminal can be adapted to be coupled to a battery (e.g., battery 102). The first voltage can be an anode voltage (V_A).

At step 604, controller circuit 400 can receive, via a second terminal (e.g., terminal 234), a second voltage. The second terminal can be adapted to be coupled to the transistor's second current terminal. The second current terminal can be a cathode of the diode. The second current terminal can be a drain if the transistor is NFET, and can be a source if the transistor is PFET. The second current terminal can be adapted to be coupled to a load (e.g., load 106). The second voltage can be cathode voltage (V_C).

At step 606, based on the first voltage and the second voltage, controller circuit 400 can close a switch (e.g., switch 406) between a voltage reference (e.g., voltage reference 408) and a third terminal (e.g., terminal 232) of the controller circuit coupled to a gate of the transistor, in which a voltage of the gate is set by the voltage reference to form a conduction channel between the first current terminal and the second current terminal.

Specifically, controller circuit 400 can include switch 406 coupled between terminal 232 (and gate 204 of transistor 202) and voltage reference 408. Voltage reference 408 can provide a target gate voltage to turn on/enable transistor 202. The target gate voltage can be higher than the source voltage of transistor 202 if transistor 202 is NFET. The target gate voltage can also be lower than the source voltage of transistor 202 if transistor 202 is PFET. Voltage reference 408 can be provided by a local voltage generator circuit, such as local voltage generator circuit 256, based on the anode voltage.

If the anode-cathode voltage V_{AC} exceeds reverse bias threshold voltage V_R and forward conduction threshold voltage V_{F-on} , RCB logic 264 can generate control signal 280a to open switch 266. Based on control signal 280a, control circuit 400 can close switch 406 to connect the gate of transistor 202 to voltage reference 408 to raise (or reduce) the gate voltage. In some examples, controller circuit 400 can generate a pulse having a pre-determined pulse width to

17

close switch **406**, with the start of the pulse triggered by control signal **280a**. In some examples, control circuit **400** can end the pulse (and open switch **406**) based on the gate-source voltage V_{GS} (or source-gate voltage V_{SG} for PFET) exceeding the threshold voltage V_{th} of transistor **202** for forming the conduction channel between first current terminal **206** and second current terminal **208**.

Step **606** can be performed as part of a rectification operation during the positive half-cycle of the AC signals. In some examples, if the positive half-cycle has not yet ended when the pulse ends and switch **406** is opened, controller circuit **400** can close switch **274** to enable amplifier **272** to further adjust the gate voltage to regulate V_{AC} at the target forward voltage V_{F-reg} . If the positive half-cycle has ended and the system is in the negative half-cycle of the AC signals, RCB logic **264** can close switch **266** to set the V_{GS} voltage (or V_{SG} voltage for PFET) to zero to disable transistor **202**.

Accordingly, in some examples as described above, a protection system can include a controller circuit and a transistor, which can be a power transistor. Examples of protection system **100** are shown in FIG. 1 through FIG. 4B, whereas examples of controller circuit **400** are shown in FIGS. 4A through 4D. Referring to FIG. 4A, controller circuit **400** can include a first terminal (e.g., terminal **230**), a second terminal (e.g., terminal **234**), and a third terminal (e.g., terminal **232**). The first terminal is adapted to be coupled to a first current terminal of a transistor (e.g., first current terminal **206** of transistor **202**). The second terminal is adapted to be coupled to a second current terminal of the transistor (e.g., second current terminal **206** of transistor **202**). The third terminal is adapted to be coupled to a gate of the transistor (e.g., gate **204**). Controller circuit **400** can control the transistor to emulate an ideal diode, of which the first current terminal is the anode and the second current terminal is the cathode. The anode can be adapted to be coupled to a battery (e.g., battery **102**), and the cathode can be adapted to be coupled to a load (e.g., load **106**). In a case where the transistor is an NFET, the first current terminal can be a source and the second current terminal can be a drain. In a case where the transistor is a PFET, the first current terminal can be a drain and the second current terminal can be a source.

Referring to FIG. 4A, controller circuit **400** can include gate control circuit **410**, which can include reverse current blocking (RCB) circuit **260** and forward conduction (FC) control circuit **402**. FC control circuit **402** can include a FC acceleration circuit **404** as well as amplifier **272**. Each of RCB circuit **260**, FC acceleration circuit **404**, and amplifier **272** can be coupled to terminal **232** (and gate **204** of transistor **202**) to set a gate voltage. RCB circuit **260** can set the gate-source voltage V_{GS} (or source-gate voltage V_{SG} for PFET) to zero to disable transistor **202** if the anode-cathode voltage V_{AC} is below a reverse bias threshold voltage V_R or a forward conduction threshold voltage V_{F-on} . RCB circuit **260** may include switch **266** coupled between terminals **230** and **232**, and may generate control signal **280a** to close switch **266** to set V_{GS}/V_{SG} to zero to disable transistor **202**, or to open switch **266** to enable FC control circuit **402** to set the gate voltage. In some examples, RCB circuit **260** can disable transistor **202** in the negative half-cycles of the AC signals that superimpose with the DC signal output by the battery, as part of a rectification operation performed by controller circuit **400** with transistor **202**.

In addition, FC control circuit **402** can enable transistor **202** by raising (or reducing) the gate voltage to increase V_{GS}/V_{SG} to enable transistor **202**, if V_{AC} is above V_{F-on} . FC

18

acceleration circuit **404** can include switch **406** coupled between voltage reference **408** and terminal **232**. If V_{AC} is above V_{F-on} , FC acceleration circuit **404** can close switch **406** to connect terminal **232** (and gate **204**) to voltage reference **408** to change the gate voltage. The rate at which the gate voltage changes can be based on, for example, the on-resistance of switch **406**, the capacitance of gate **204**, etc., and can be independent from the gain/transconductance of amplifier **272**. FC acceleration circuit **404** can generate a pulse to close switch **406** for a finite duration. The start time of the pulse can be based on detection of V_{AC} being above V_{F-on} (e.g., based on control signal **280a**). The end time of the pulse can be based on a pre-determined delay from the start time, or based on the V_{GS} voltage (or V_{SG} voltage for PFET) exceeding a threshold voltage (e.g., V_{th} of transistor for forming the conduction channel). In some examples, FC control circuit **402** can enable transistor **202** in the positive half-cycles of the AC signals, as part of a rectification operation performed by controller circuit **400** with transistor **202**.

Moreover, the output of amplifier **272** can be coupled to terminal **232** via switch **274**. After the pulse that closes switch **406** ends, if V_{AC} is still above V_{F-on} (e.g., the positive half-cycle has not yet ended), FC control circuit **402** can close switch **274** to allow amplifier **272** to further adjust the gate voltage by regulating V_{AC} at the target forward voltage V_{F-reg} .

In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action, then: (a) in a first example, device A is coupled to device B; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not substantially alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A. Also, in this description, a device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof. Furthermore, in this description, a circuit or device that includes certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, such as by an end-user and/or a third party.

While certain components may be described herein as being of a particular process technology, these components may be exchanged for components of other process technologies. Circuits described herein are reconfigurable to include the replaced components to provide functionality at

19

least partially similar to functionality available before the component replacement. Components illustrated as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the illustrated resistor. For example, a resistor or capacitor instead be multiple resistors or capacitors, respectively, coupled in series or in parallel between the same two nodes as the single resistor or capacitor. Also, uses of the phrase “ground voltage potential” in this description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. Unless otherwise stated, “about,” “approximately,” “nearly,” or “substantially” preceding a value means ± 10 percent of the stated value.

Modifications are possible in the described examples, and other examples are possible, within the scope of the claims. What is claimed is:

1. A system comprising:

a transistor including a control terminal, a first terminal, a second terminal; and

a control circuit including:

a first circuit including:

a first comparator including a first input, a second input, and an output;

a second comparator including a first input, a second input, and an output; and

a logic circuit coupled to the output of the first comparator and the output of the second comparator, the logic circuit including a first control output; and

a second circuit including:

a second control output coupled to the control terminal of the transistor; and

a third circuit including an input, a first output, and a second output, wherein the input of the third circuit is coupled to the first control output, and the first output of the third circuit is coupled to the second control output.

2. The system of claim 1, wherein:

the second circuit includes:

a first switch including a first terminal coupled to the second control output;

a third control output coupled to the first switch; and an amplifier including a first input, a second input, and an output coupled to a second terminal of the first switch.

3. The system of claim 2, wherein:

a first reference is a target forward voltage.

4. The system of claim 2, wherein:

the first circuit includes a second switch coupled to the first control output;

a first terminal of the second switch is coupled to the second control output;

a second terminal of the second switch is coupled to the first terminal of the transistor; and

the third circuit includes a third switch coupled between a first voltage and the second control output.

5. The system of claim 4, wherein:

the first comparator is coupled to a second voltage and a second reference; and

in response to first comparator determining the second voltage is less than the second reference, the third circuit is configured to open the third switch.

20

6. The system of claim 4, wherein:

the second comparator is coupled to a second voltage and a third reference; and

in response to the second comparator determining the second voltage is greater than the third reference, the third circuit is configured to close the third switch.

7. The system of claim 4, wherein:

the control circuit is configured to determine a positive half-cycle of a second voltage, the second voltage being coupled to the second input of the first comparator and the second input of the second comparator; and in response to determining the positive half-cycle of the second voltage is not ended, the third circuit is configured to output a signal on the third control output to close the first switch.

8. The system of claim 7, wherein:

in response to determining the positive half-cycle of the second voltage has ended, the third circuit is configured to open the third switch.

9. The system of claim 1, wherein:

a first voltage, coupled to the second input of the first comparator and coupled to the second input of the second comparator, is a difference voltage between a second voltage at the first terminal of the transistor and a third voltage at the second terminal of the transistor;

a first reference, coupled to the first input of the first comparator, is a reverse bias threshold voltage of a body diode of the transistor; and

a second reference, coupled to the first input of the second comparator, is forward conduction threshold voltage.

10. The system of claim 1, wherein:

the third circuit includes:

a start pulse circuit including an input and an output, the input of the start pulse circuit coupled to the first control output;

an end pulse circuit including an input and an output, the input of the end pulse circuit coupled to the output of the start pulse circuit; and

a pulse generator circuit including a first input coupled to the output of the start pulse circuit, a second input coupled to the output of the end pulse circuit, and an output.

11. The system of claim 10, wherein:

the start pulse circuit includes an edge detection circuit; and

the end pulse circuit include a delay circuit.

12. A controller circuit comprising:

a first circuit including:

a first comparator including a first input, a second input, and an output;

a second comparator including a first input, a second input, and an output; and

a logic circuit coupled to the output of the first comparator and the output of the second comparator, the logic circuit including a first control output; and

a second circuit including:

a second control output; and

a third circuit including an input, a first output, and a second output, wherein the input of the third circuit is coupled to the first control output, and the first output of the third circuit is coupled to the second control output.

13. The controller circuit of claim 12, wherein:

the second circuit includes:

a first switch including a first terminal coupled to the second control output;

a third control output coupled to the first switch; and

21

an amplifier including a first input, a second input, and an output coupled to a second terminal of the first switch.

14. The controller circuit of claim 13, wherein:
the first circuit includes a second switch coupled to the first control output;
a first terminal of the second switch is coupled to the second control output; and
the third circuit includes a third switch coupled between a first voltage and the second control output.
15. The controller circuit of claim 14, wherein:
the first comparator is coupled to a second voltage and a first reference; and
in response to the first comparator determining the second voltage is less than the first reference, the third circuit is configured to open the third switch.
16. The controller circuit of claim 14, wherein:
the second comparator is coupled to a second voltage and a second reference; and
in response to the second comparator determining the first voltage is greater than the second reference, the third circuit is configured to close the third switch.
17. The controller circuit of claim 14, wherein:
the control circuit is configured to determine a positive half-cycle of a second voltage, the second voltage is coupled to the second input of the first comparator and the second input of the second comparator; and

22

in response to determining the positive half-cycle of the second voltage is not ended, the third circuit is configured to output a signal on the third control output to close the first switch.

18. The controller circuit of claim 17, wherein:
in response to determining the positive half-cycle of the second voltage has ended, the third circuit is configured to open the third switch.

19. A method comprising:
receiving, by a controller circuit, a first voltage;
receiving, by the controller circuit, a first reference;
in response to the first voltage being greater than the first reference, closing, by the controller circuit, a first switch between a second reference and a control output adapted to be coupled to a gate of a transistor; and
in response to the first voltage being less than a third reference, opening, by the controller circuit, a second switch between an amplifier and the control output.

20. The method of claim 19, further comprising:
determining, by the controller circuit, a positive half-cycle of the first voltage;
in response to determining the positive half-cycle of the first voltage is not ended, initiating, by the controller circuit, a regulation loop configured to regulate the first voltage; and
in response to determining the positive half-cycle of the first voltage has ended, opening, by the controller circuit, the second switch.

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